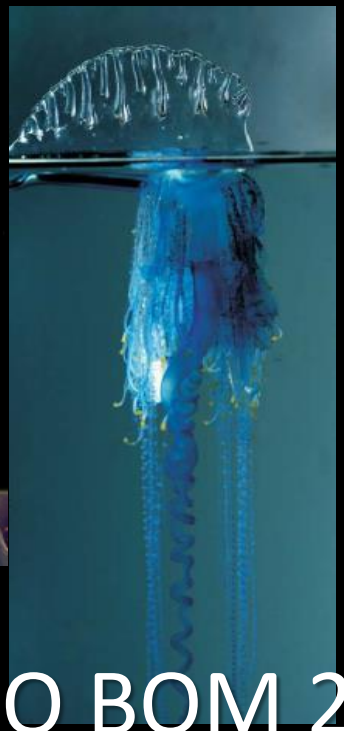
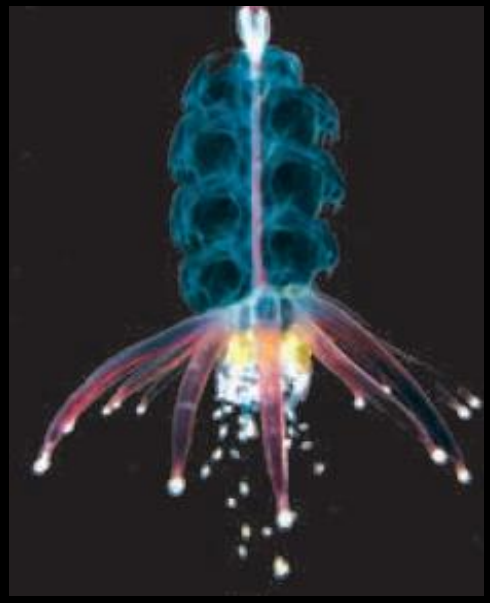
PHYLUM CNIDARIA









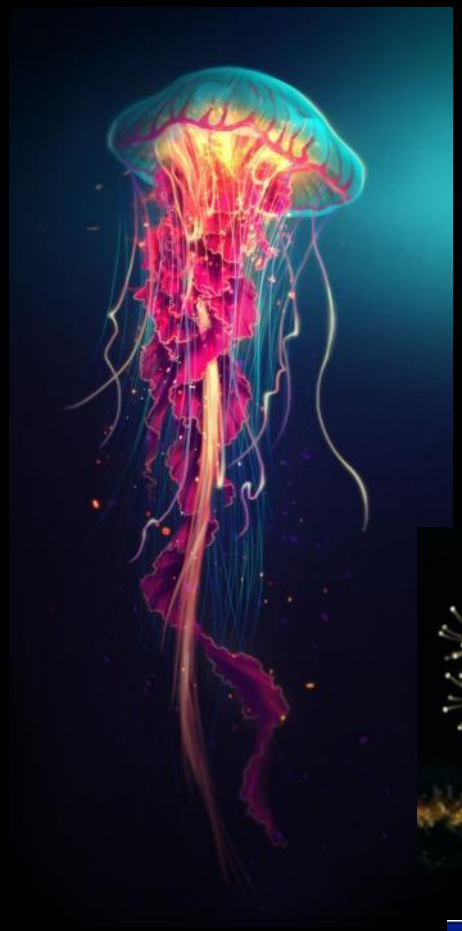

CURSO BOM 2026

Biología de los Organismos Marinos:

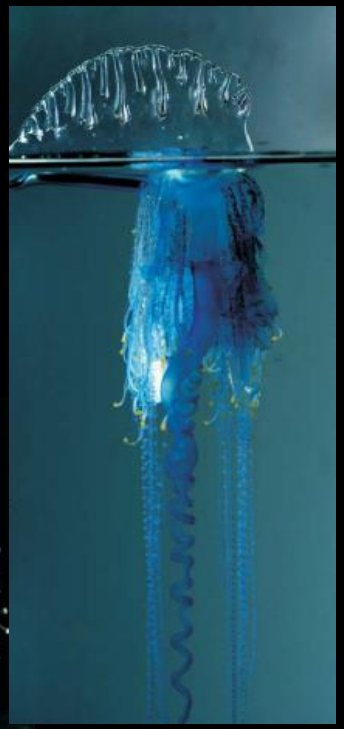
Foraminíferos e Invertebrados

GABRIELA FAILLA SIQUIER
Sección Zoología de Invertebrados
Dpto. de Biología Animal

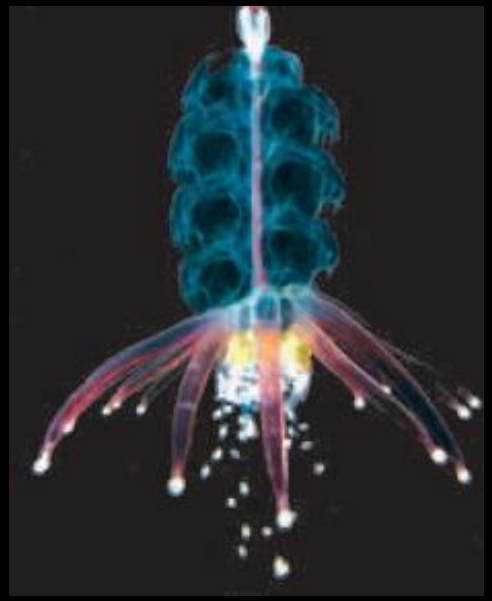
PHYLUM CNIDARIA



















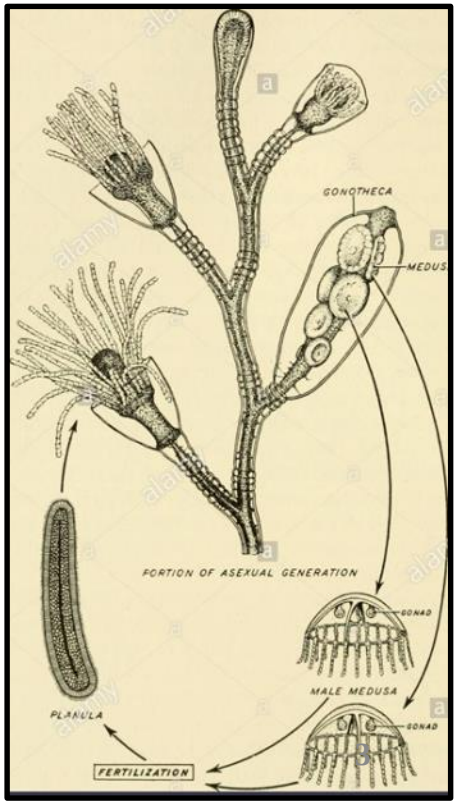


PHYLUM CNIDARIA

(gr. Knide: ortiga)

- **CNIDOCITOS o CNIDAE**
- **METAGÉNESIS pólipo - medusa**
  asexual - sexual

- **Metazoa diblásticos ectodermo/endodermo**
- **Mesoglea**
- **Nivel de organización tisular**
- **Simetría radial (bi o tetrarradial)**
- **Larva plánula**
- **Predadores**
- **Marinos *(pocas sp. dulceacuícolas)***



PHYLUM CNIDARIA

(Kramp 1961, Bouillion, 1999)

SUBPHYLUM MEDUSOZOA



1. **CLASE HYDROZOA (pm)**
2. **CLASE SCYPHOZOA (pM)**
3. **CLASE CUBOZOA (pM)**
4. **CLASE STAUROZOA (pM)**

SUBPHYLUM ANTHOZOA



1. **CLASE ANTHOZOA (P)**
   Subclase:
   i. **Hexacorallia**
   ii. **Octocorallia**
   iii. **Ceriantharia**

PHYLUM CNIDARIA (19 579)

(Kramp 1961, Bouillion, 1999)

1. SUBPHYLUM MEDUSOZOA



1. **CLASE HYDROZOA (pm)**
2. **CLASE SCYPHOZOA (pM)**
3. **CLASE CUBOZOA (pM)**
4. **CLASE STAUROZOA (pM)**

2. SUBPHYLUM ANTHOZOA

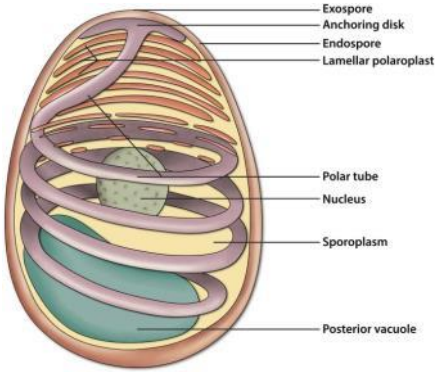


1. **CLASE ANTHOZOA (P)**

Subclase:

i. Hexacorallia
ii. Octocorallia
iii. Ceriantharia

3. SUBPHYLUM: ENDOCNIDOZOA ≈2500 spp

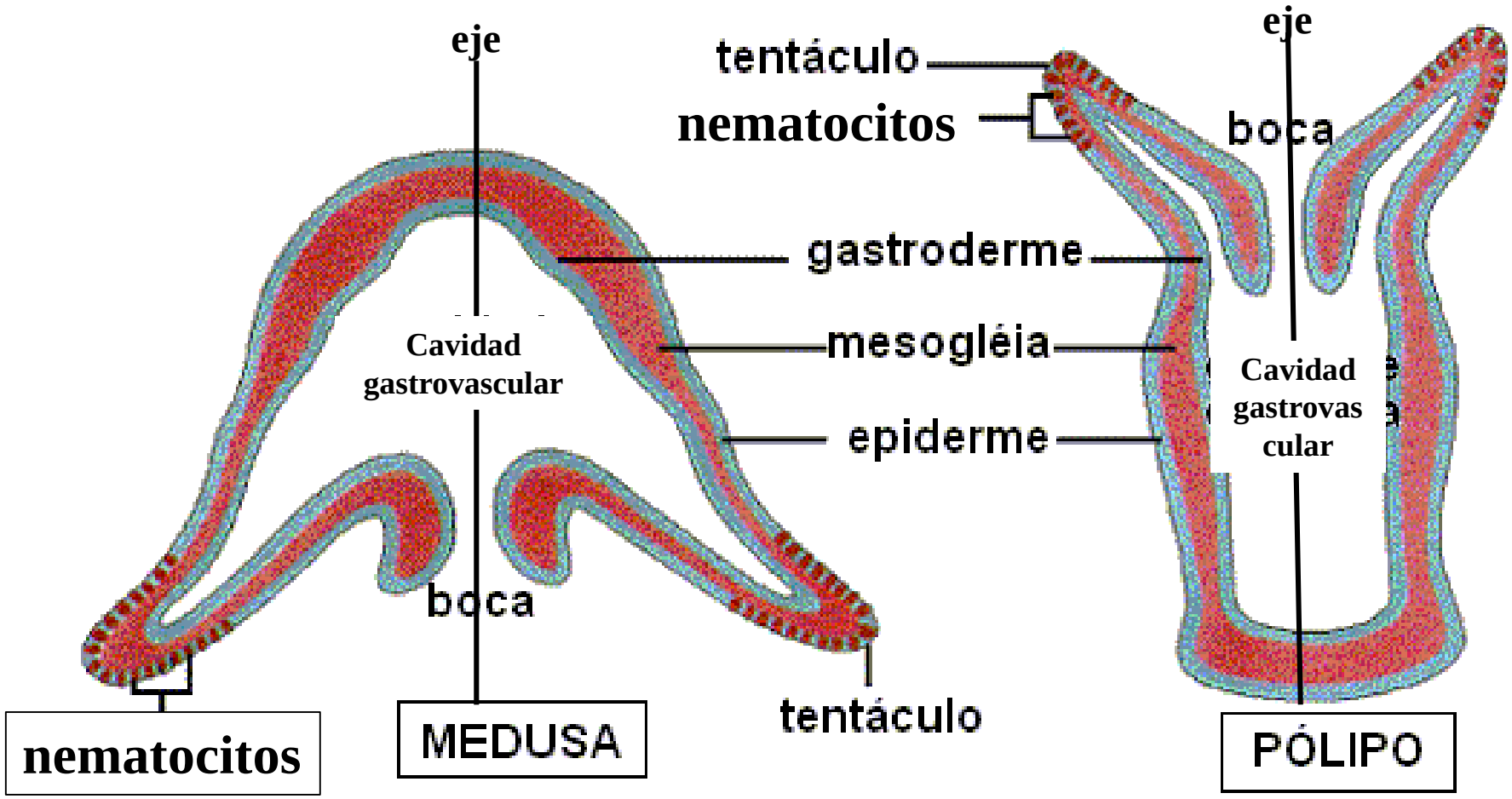


CLASE MIXOZOA

CLASE POLYPODIOZOA.

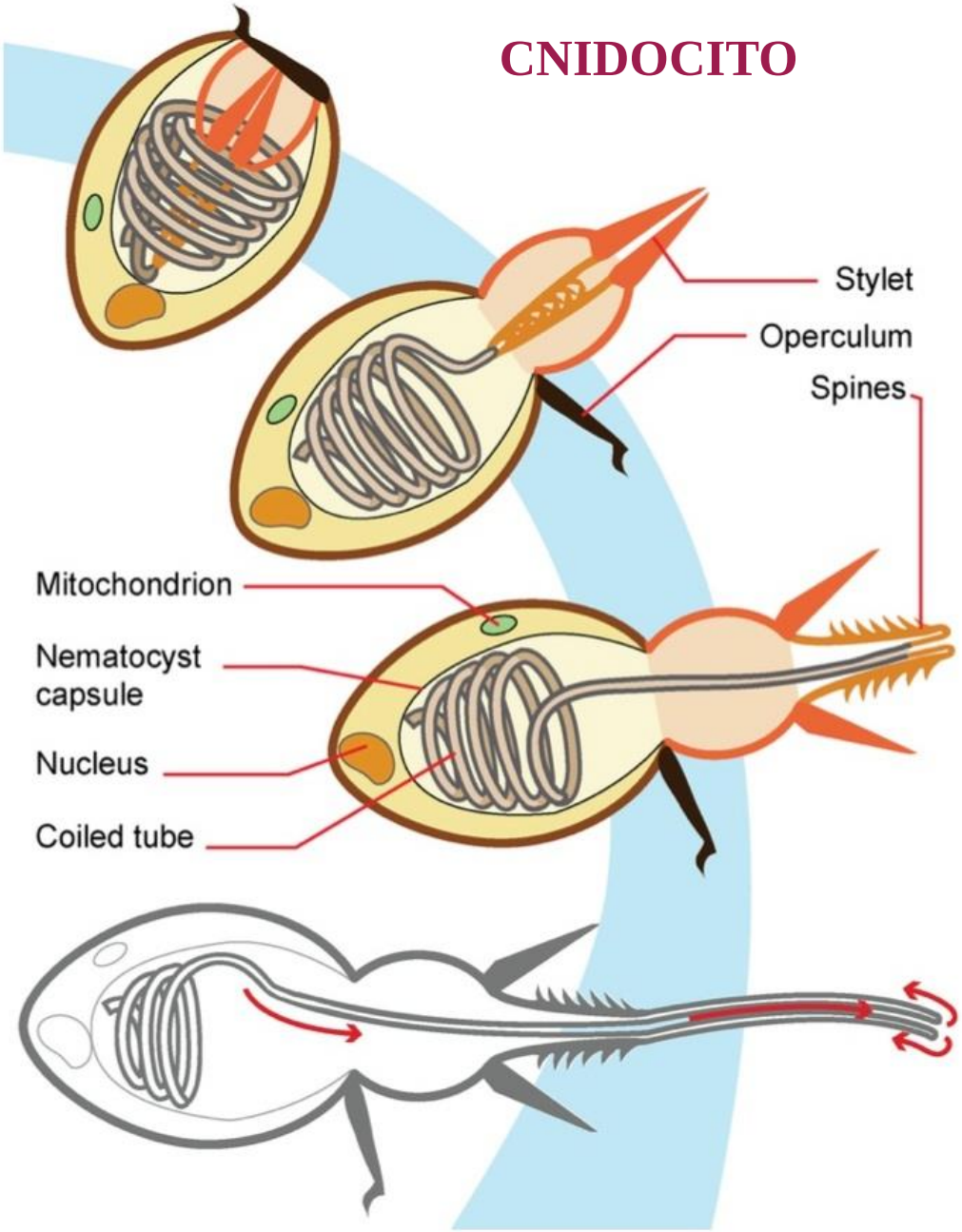
<u>Todos endoparásitos</u> de peces (anfibios y reptiles) con hospedador intermediario poliqueto

Plan básico generalizado



cara oral **cara aboral**

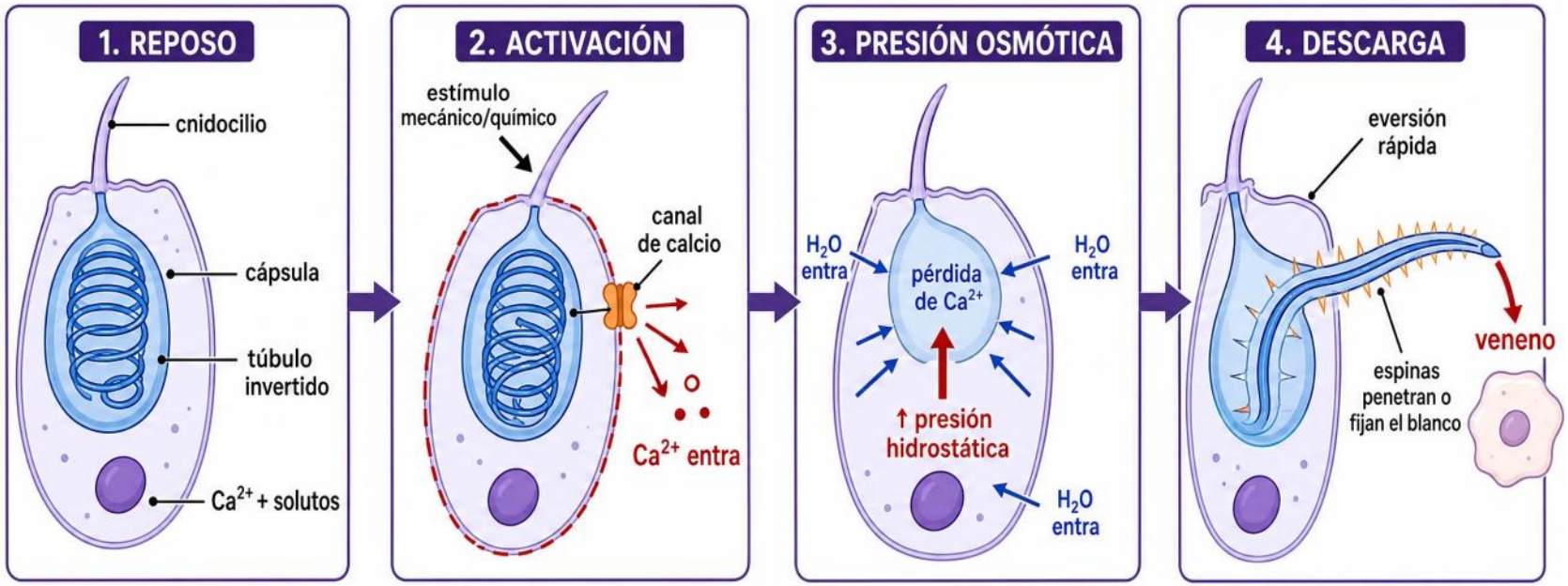
CNIDOCITO


Stylet
Operculum
Spines
Mitochondrion
Nematocyst capsule
Nucleus
Coiled tube


10 µm


cifonauta.cebimar.usp.br

Mecanismo de disparo del cnidocito



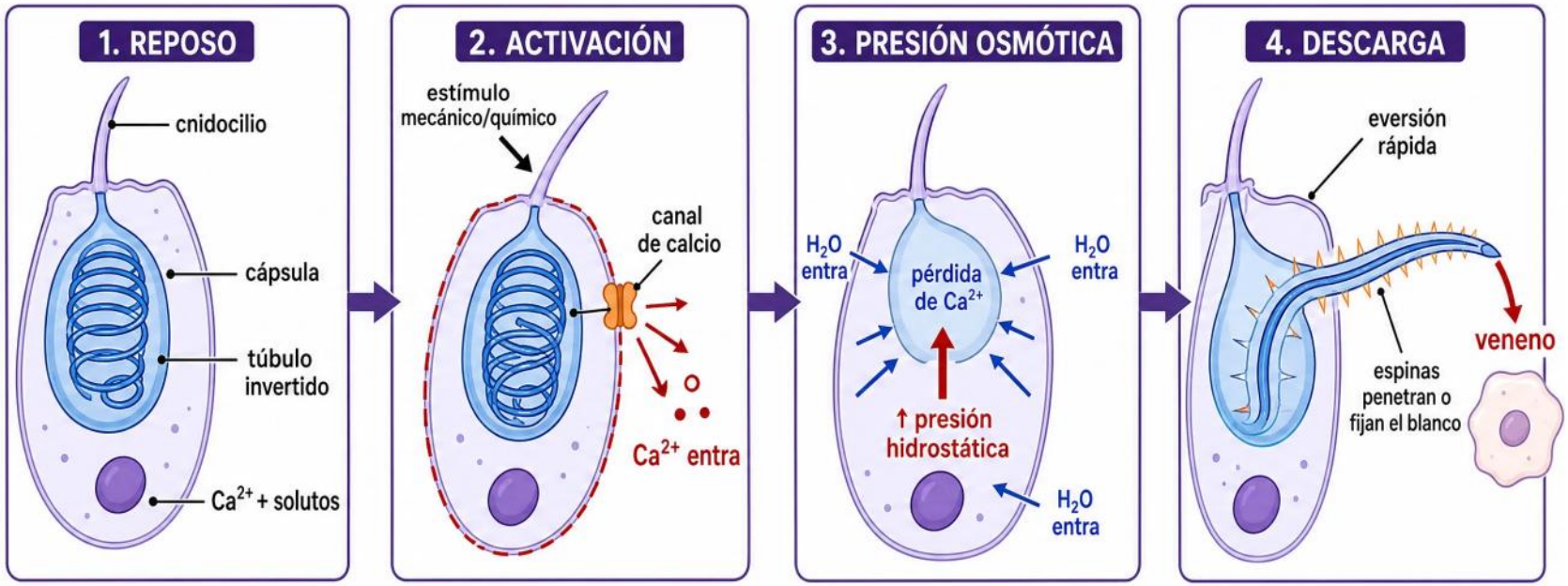
2. Ca^{2+} entra en el citoplasma del cnidocito a través de canales

→ **activación del <u>cnidocito</u>** (célula)

3. Ca^{2+} sale de la cápsula (<u>nematocisto</u>) y entra agua

→ **aumenta la presión de la cápsula**

Mecanismo de disparo del cnidocito



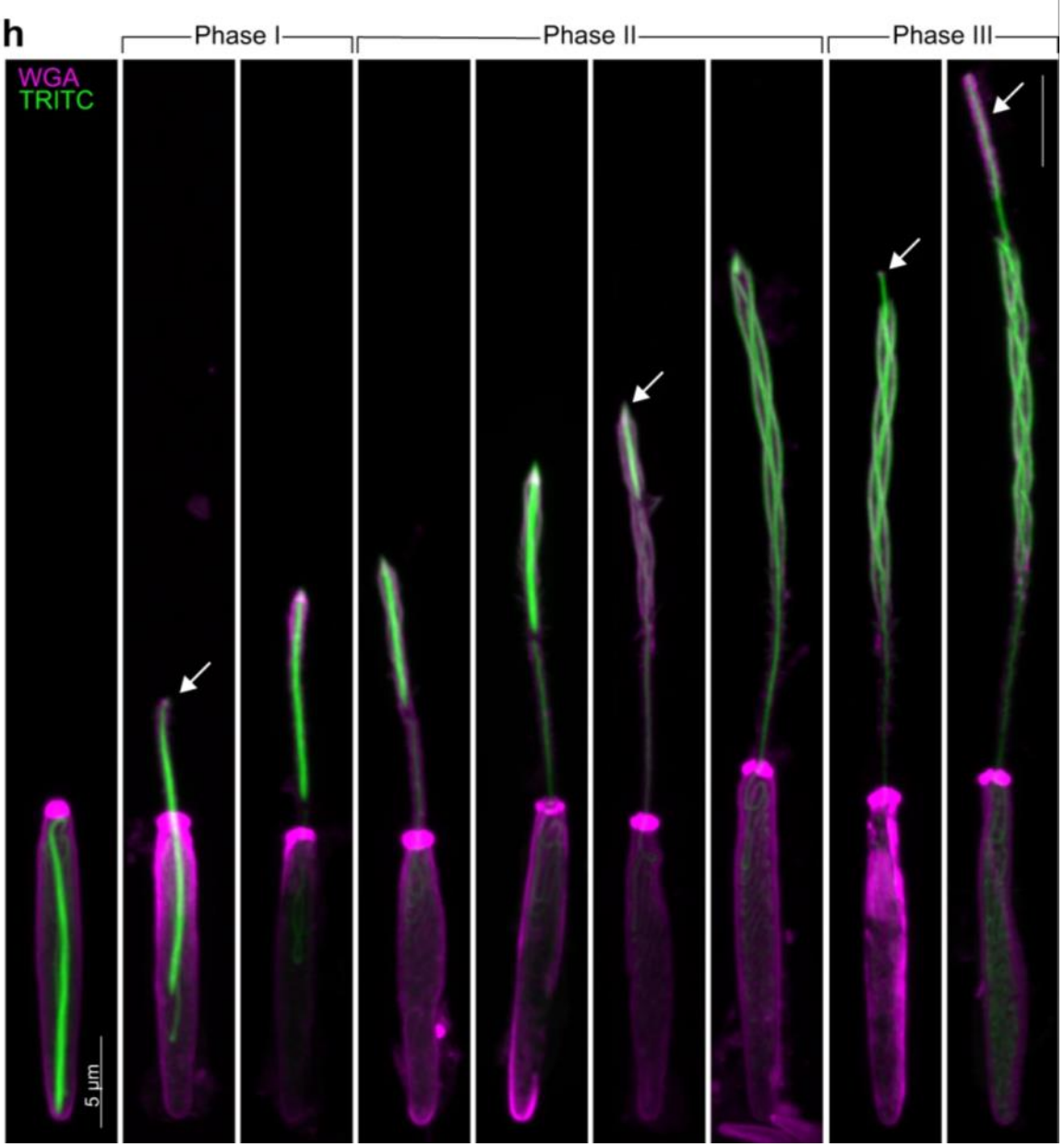
2. activación del <u>cnidocito</u> (célula)

→ **Ca^{2+} entra en el citoplasma del cnidocito a través de canales**

3. Ca^{2+} sale de la cápsula (<u>nematocisto</u>) y entra agua

→ **aumenta la presión de la cápsula**

Live imaging of nematocyst discharge



Representative images of the distinct phases of thread eversion. Fluorescent staining shows the geometric transformations in each phase. Arrows indicate the apex of the thread during distinct phases.

CNIDOCITO = CÉLULA

CNIDOCISTO = CÁPSULA (=CNIDA)

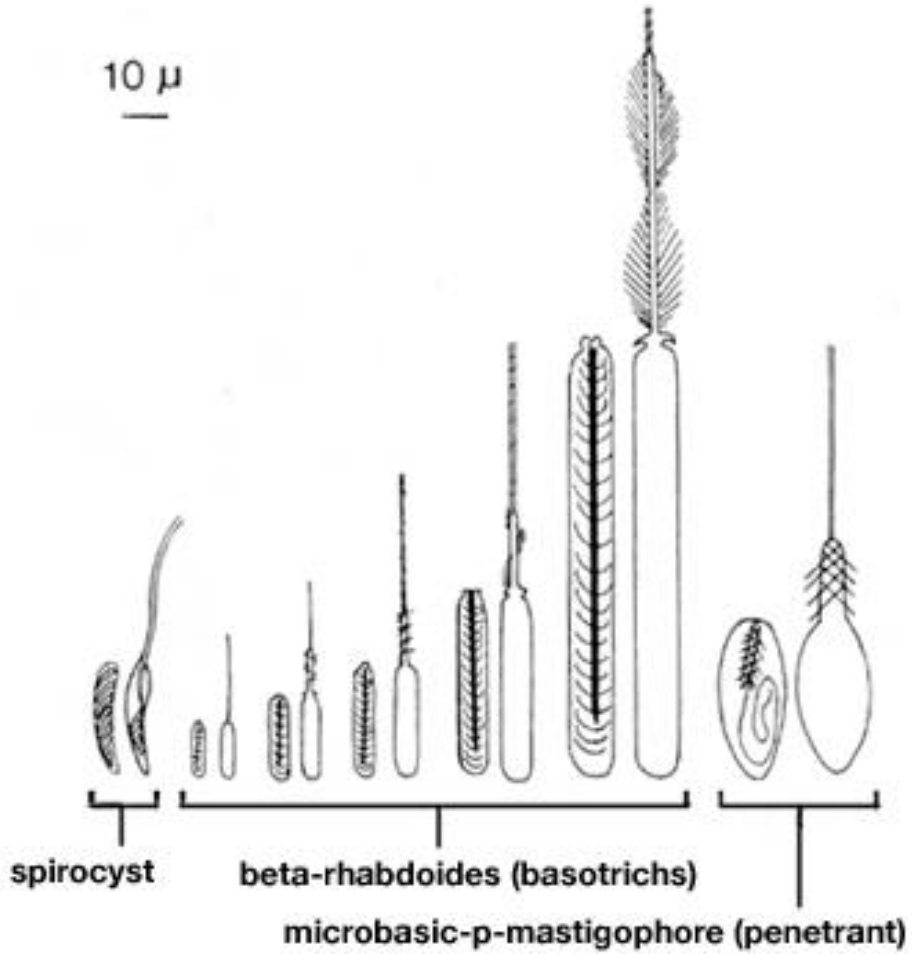
3 CATEGORÍAS DE CNIDOCISTOS:

1. **NEMATOCISTOS (TÓXICO)** (≈ **30** tipos) MEDUSOZOA Y ANTHOZOA
2. **SPIROCISTOS (NO TÓXICO) (2)**
3. **PTYCOCISTOS (NO TÓXICO) (1)**

(solo en Subclase Ceriantharia)

sólo ANTHOZOA





T Tent. faringe, mesent. T



<u>CNIDOMA</u>: Conjunto de los diferentes tipos de cnidocistos que tiene una especie

MATRIZ EXTRA CELULAR (MEC)

MESOGLEA (MEC)

FUNCIÓN (medusas):



- SOPORTE

MESOGLEA (MEC)
Sustancia con propiedades viscoelásticas

- **FIBRAS DE COLÁGENO (principal elemento de soporte)**
- **MICROFIBRILLAS (= fibrilinas- ppal. elem. flexible)**
- Otras proteínas estructurales (elastina, fibronectina, laminina).
- **Glucoproteínas: proteinas unidas a cadenas cortas de azúcares**
- **CÉLULAS LIBRES ***
- **Proteoglicanos (=glucosaminoglicanos GAGs):**

  Firmeza, elasticidad

  Transmisión de señales entre interior y exterior de cels. de la mec

  RETENCIÓN de…. **AGUA !**

SOPORTE

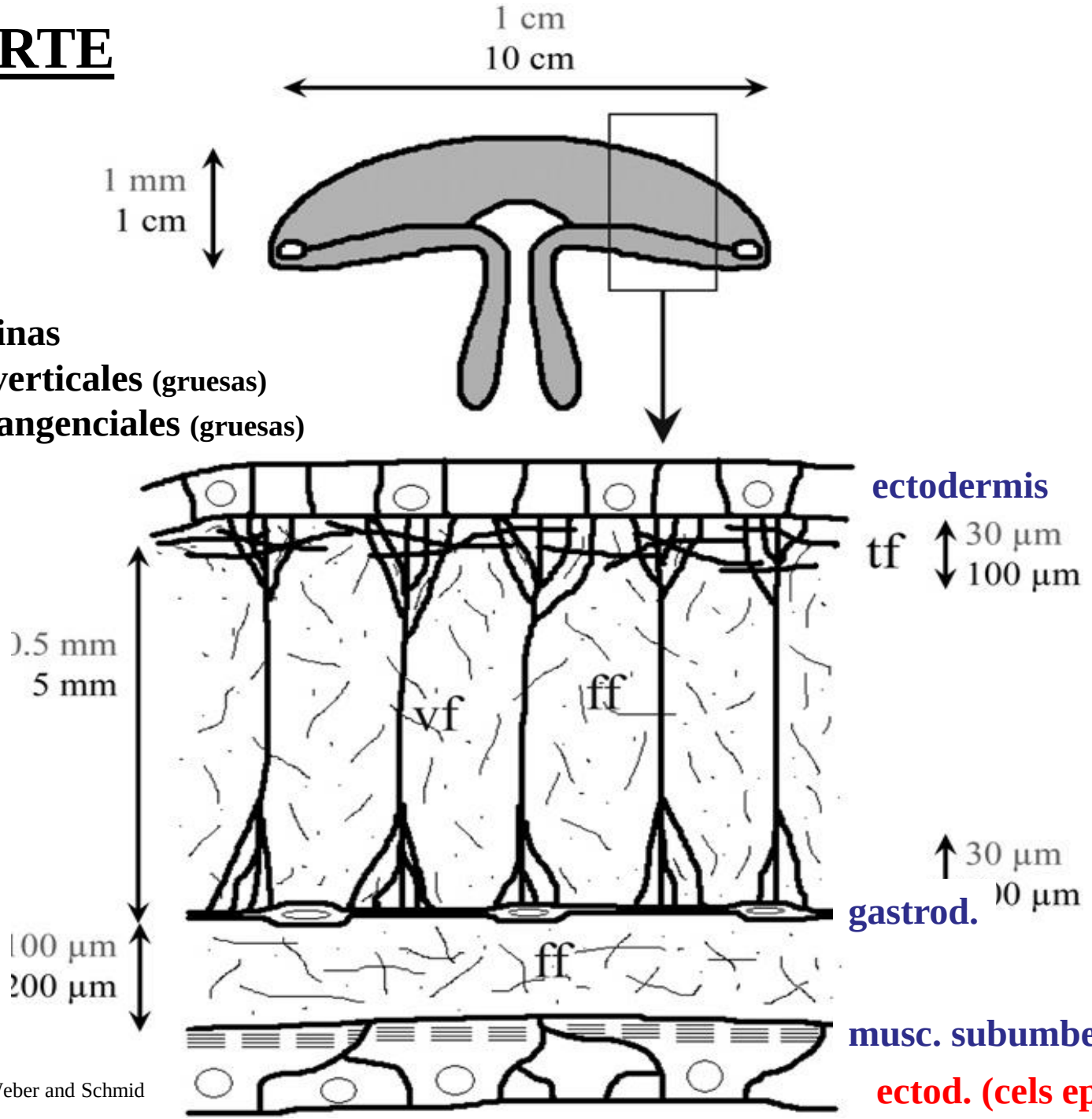
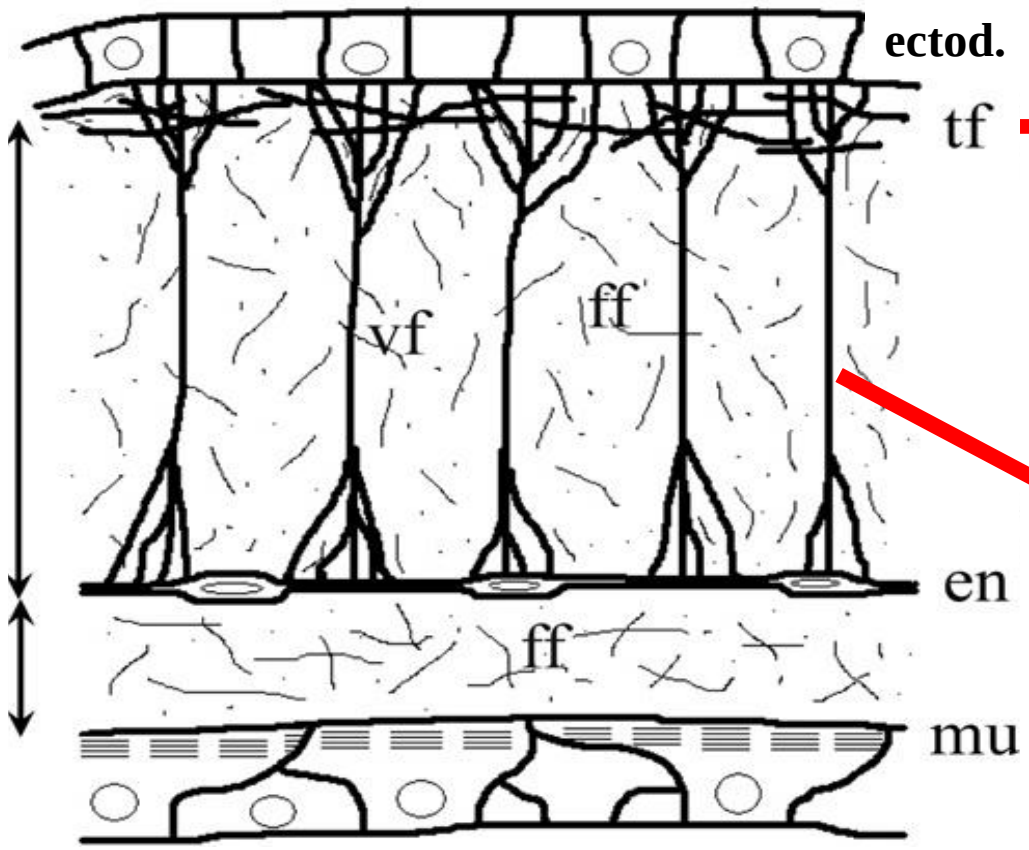
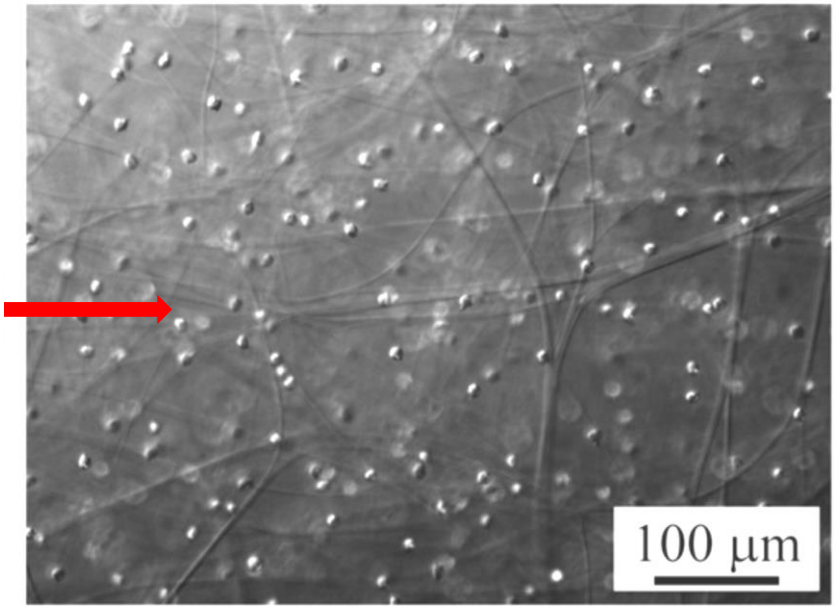
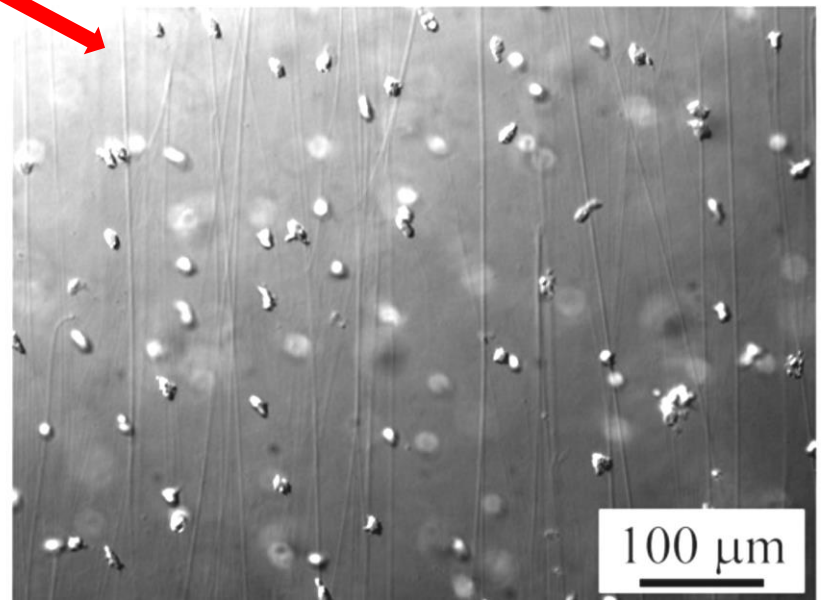


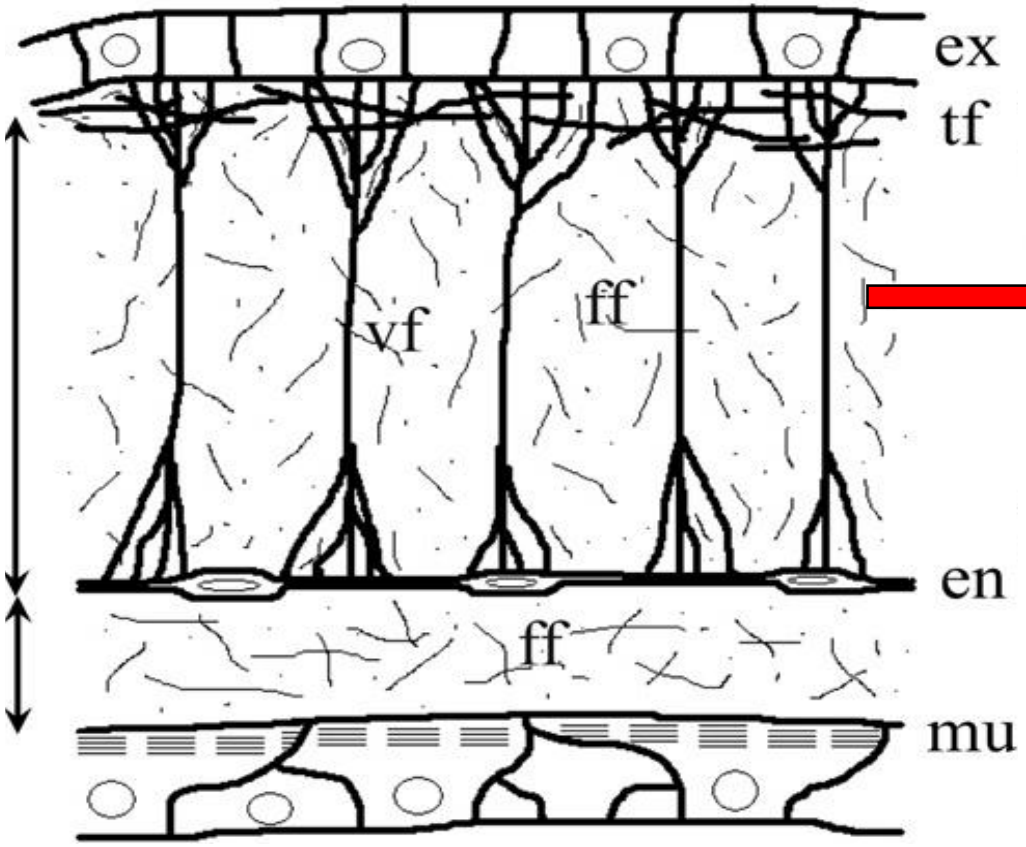
Figure adapted from Weber and Schmid

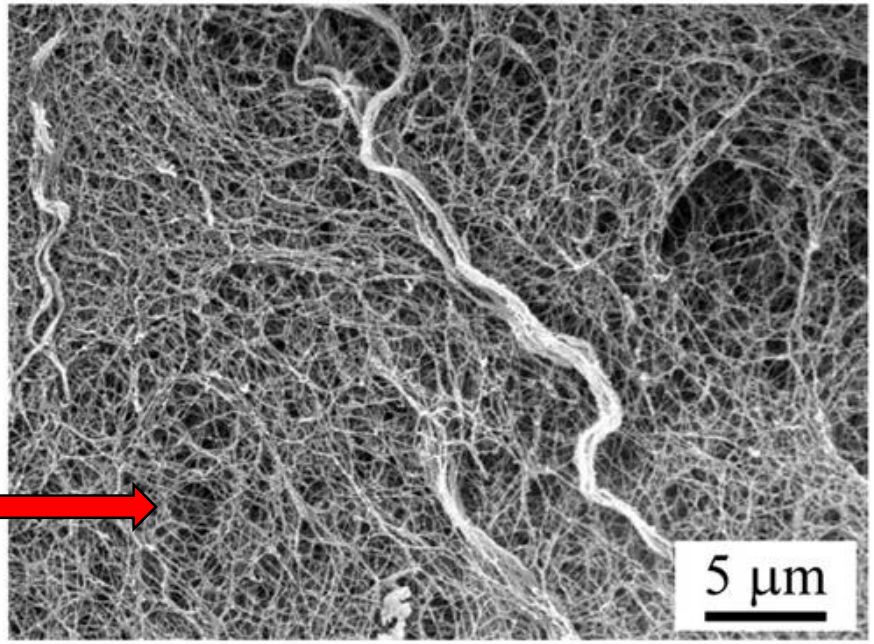


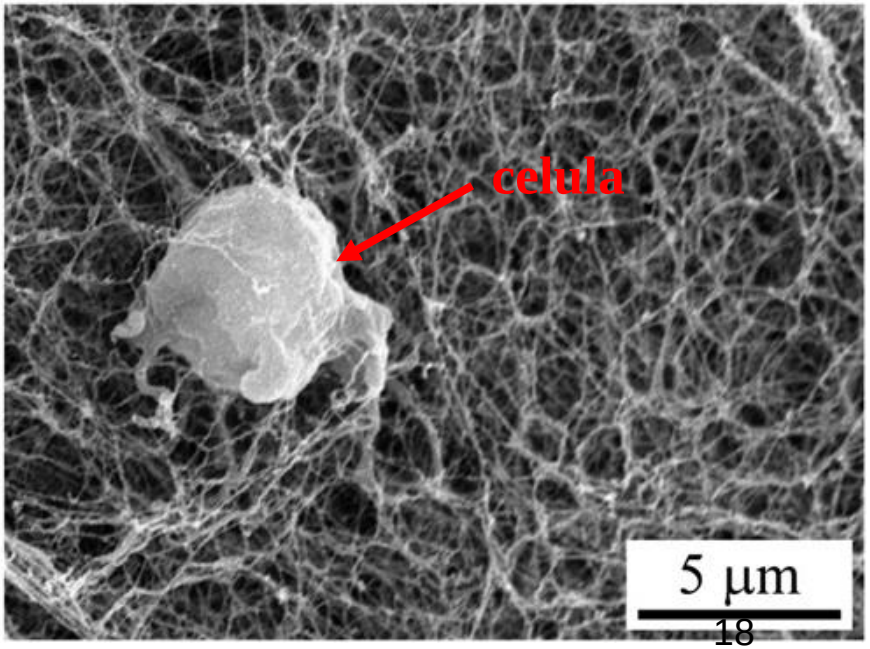




Las fibras gruesas dan soporte y rigidez







- **Las fibras finas dan flexibilidad**
(ambiente viscoso)
- **Antagonista de la contracción de las células mioepiteliales**

MESOGLEA (MEC)

FUNCIÓN (medusas):



- **SOPORTE**

- **FLOTACIÓN (intercambio iónico sulfato - cloruro)**

- **Tolerancia a la hipoxia -aporte O2 a las células por gradiente de difusión**

- **Almacenamiento y transporte de nutrientes**

- **Antagonista de la contracción muscular**

PHYLUM CNIDARIA - CICLOS DE VIDA

SUBPHYLUM MEDUSOZOA



1. CLASE HYDROZOA (pm)
2. CLASE SCYPHOZOA (pM)
3. CLASE CUBOZOA (pM)
4. CLASE STAUROZOA (pM)

SUBPHYLUM ANTHOZOA



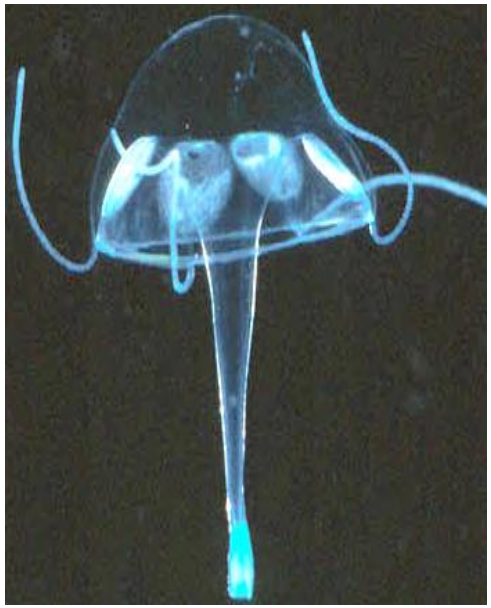
1. CLASE ANTHOZOA (P)

CLASE: HYDROZOA

≈ **3700** spp.

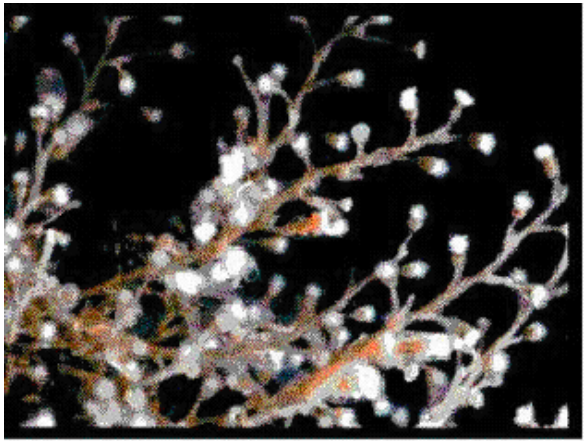


Hidromedusas

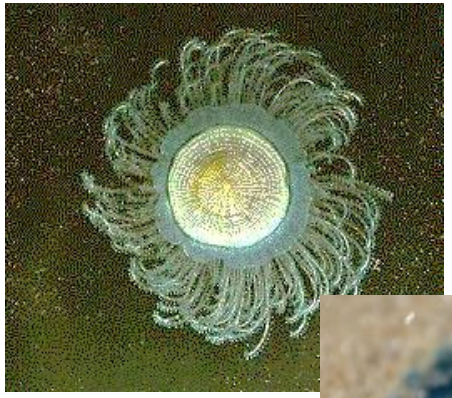




Hidrocorales



Hidroides





Condróforos

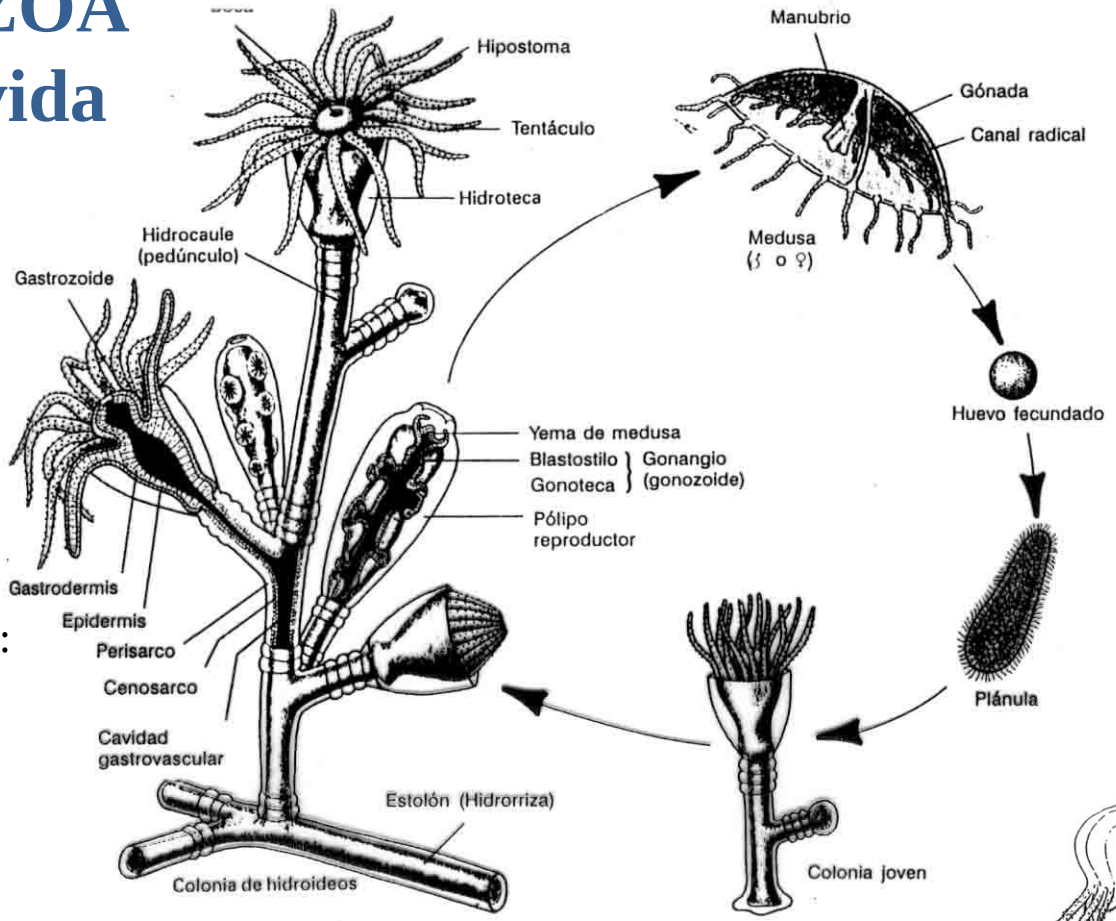


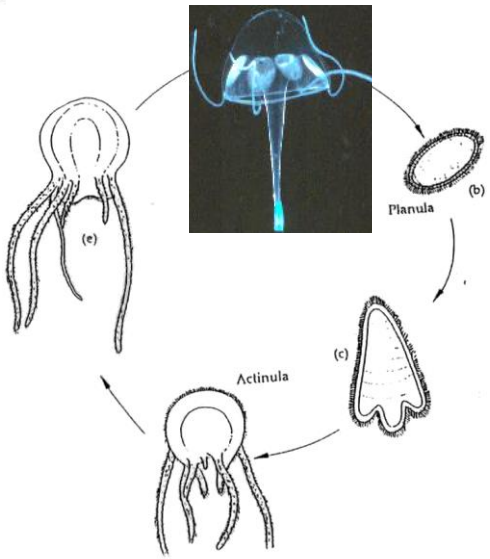
Sifonóforos

Es casi imposible generalizar un Hydrozoa típico.

HYDROZOA
ciclo de vida

colonias:
homo- heteromof.





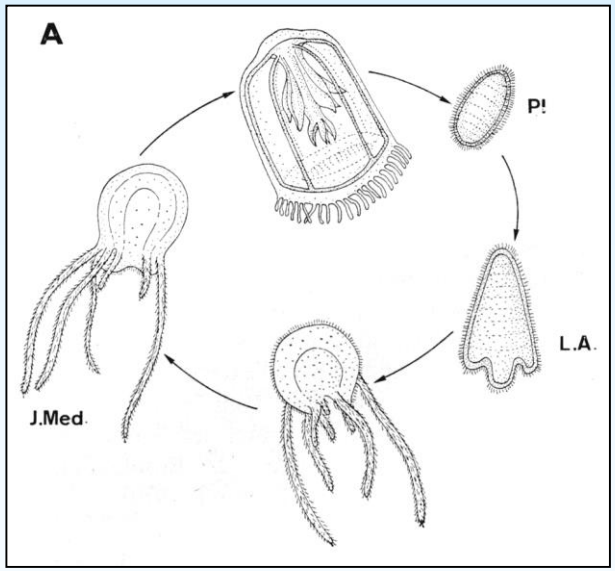
Liriope tetraphylla

Clase Hydrozoa

Subclase Trachylina

Orden:
Trachymedusa
Narcomedusa
Limnomedusa.

Pólipo - medusa



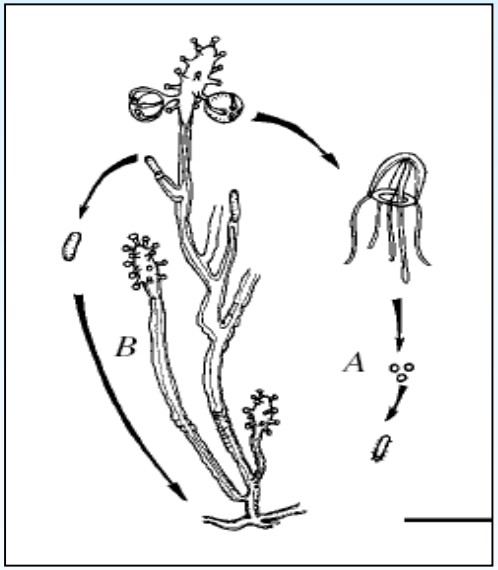
Holoplanctónicas

Sin alternancia

Subclase Hydroidolina

Orden:
Leptomedusae
Anthomedusae
Siphonophonofora

Pólipo - medusa

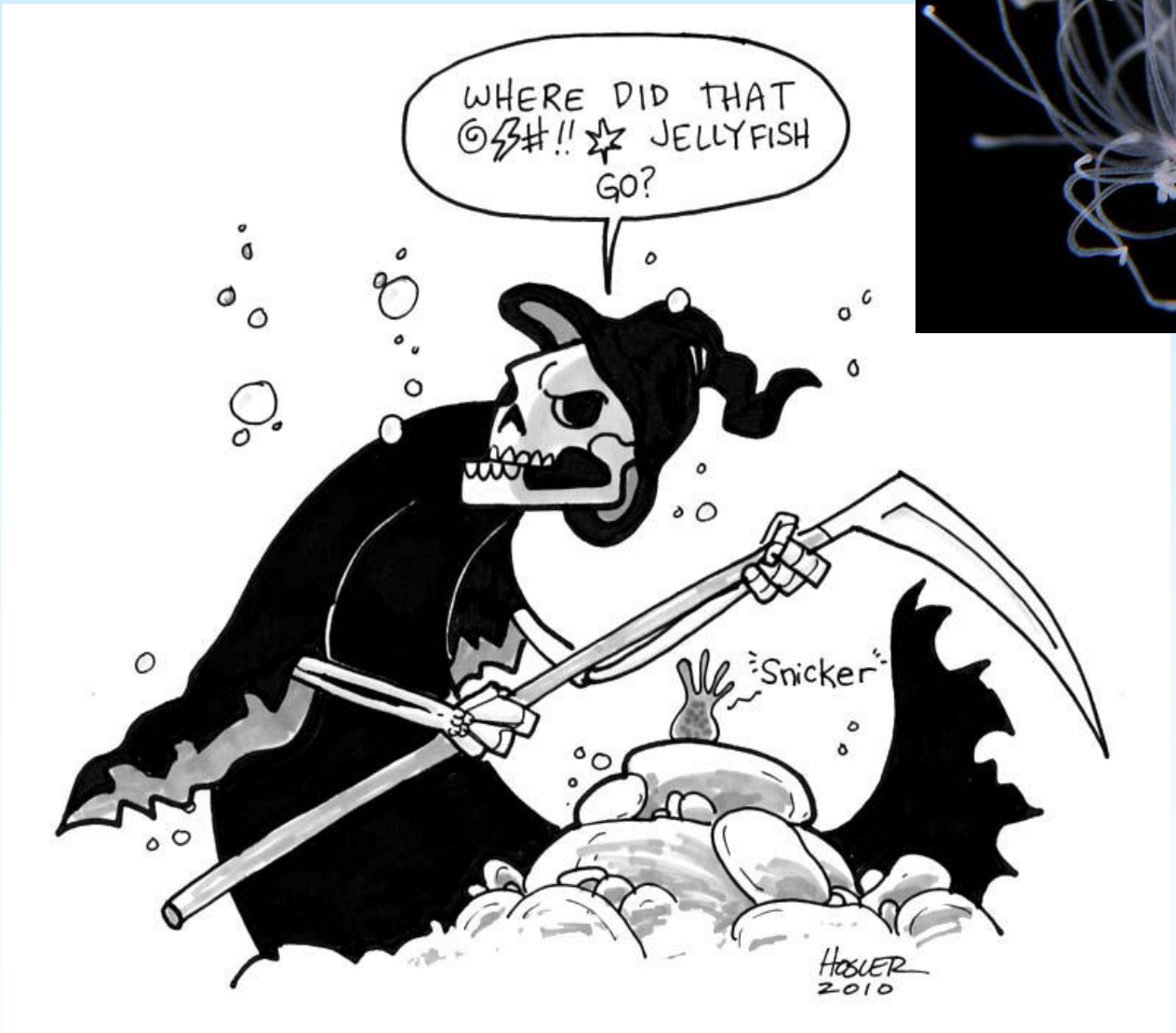


Meroplanctónicas

Con alternancia

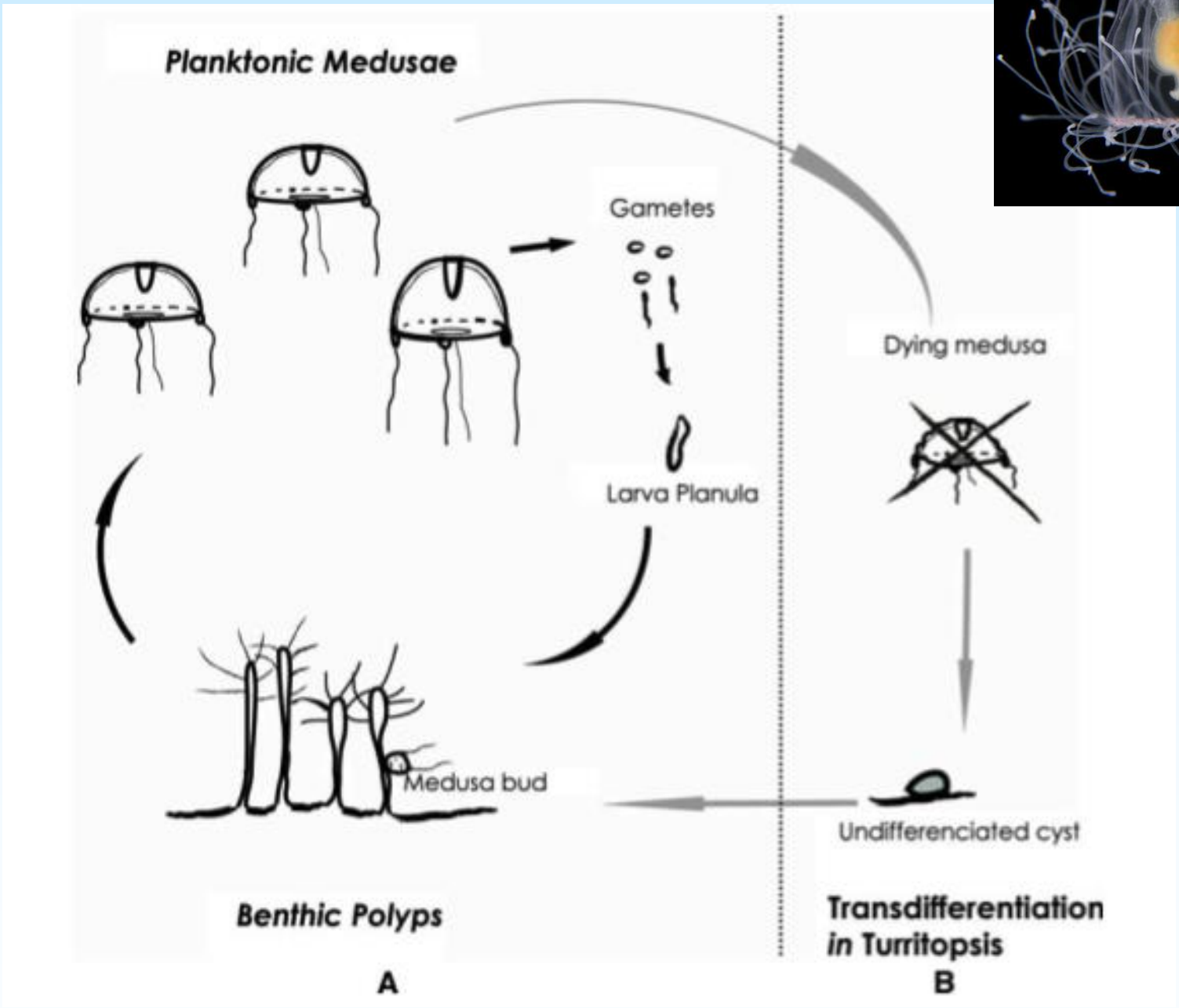
Turritopsis nutricola
La medusa inmortal





Turritopsis nutricola

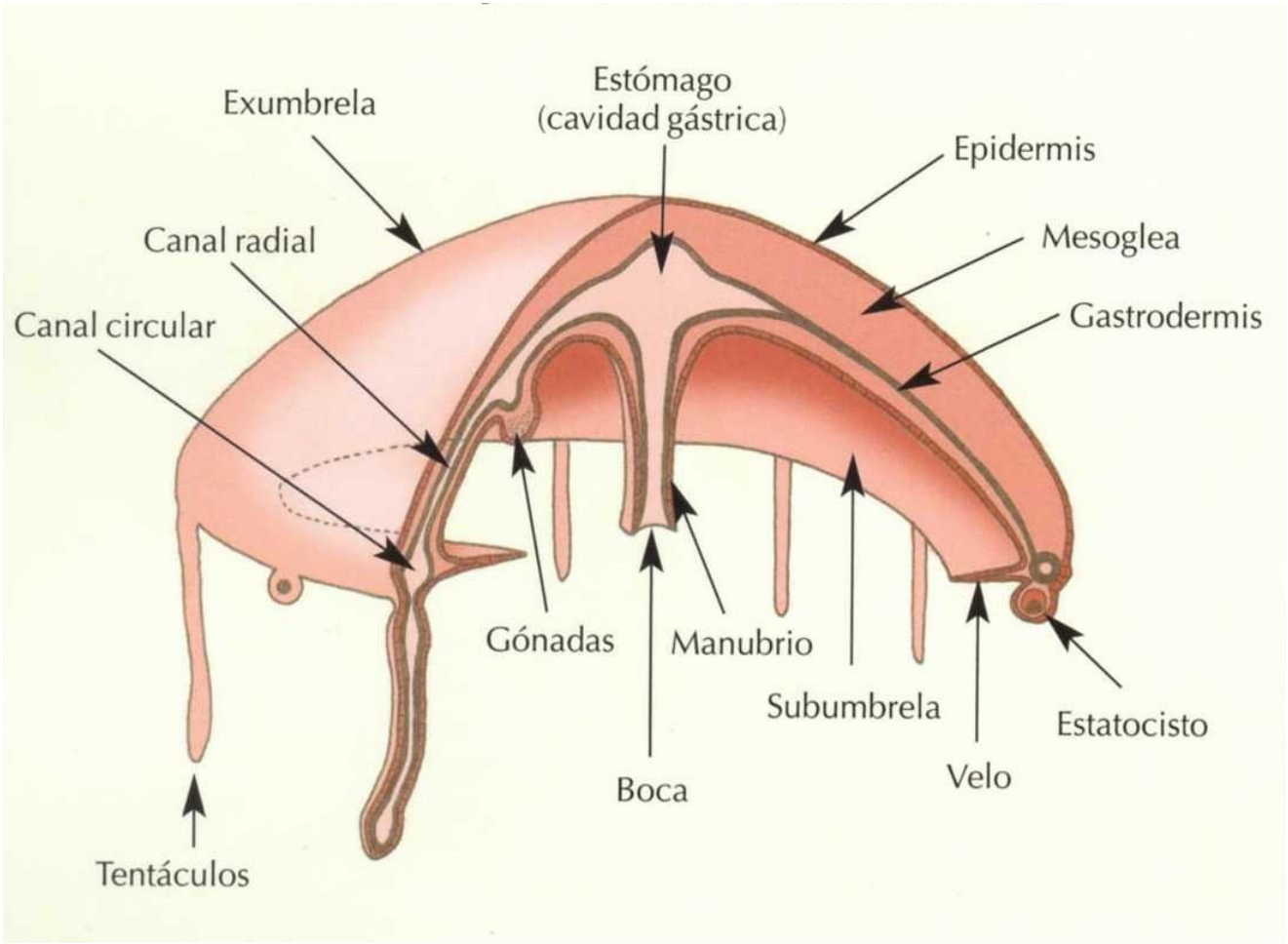




CLASE: HYDROZOA

≈ **3700** spp.

Medusa de Hydrozoa



- **No presentan brazos orales**
- **Gónada ECTODERMICA**

Subclase: Trachylina



Liriope tetraphylla (Limnomedusa)



Aequorea aequorea (Limnomedusa)

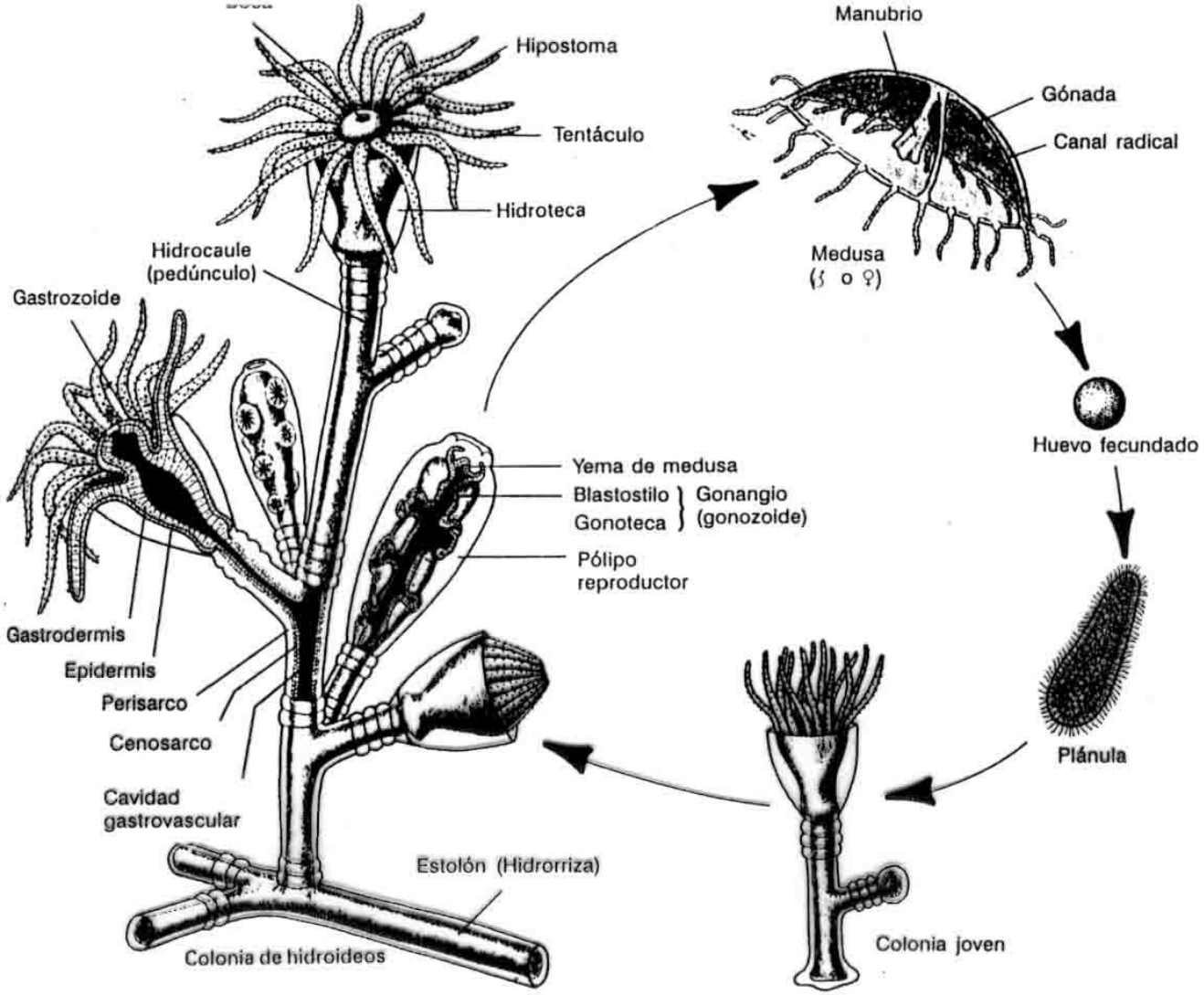


Cunina octonaria (Narcomedusa)



Olindias sambaquiensis (Limnomedusa)

HYDROZOA
ciclo de vida



colonias:
homo- heteromof.

Medusa planctónica – Pólipo bentónico

HYDROZOA
ciclo de vida

Medusa planctónica – Pólipo bentónico

EL PATRÓN NO SIEMPRE SE CUMPLE

Hydrozoa



L. tetraphylla



Cunina octonaria

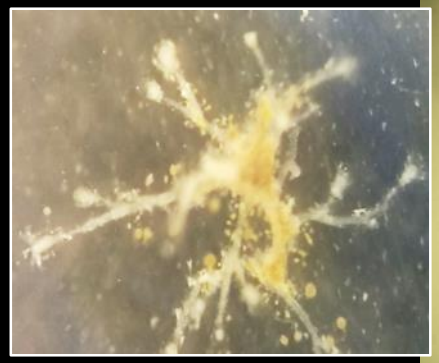


plánula de C.octonaria

Hydrozoa
SubCl:Hydroidolina



C. hemisphaerica







Pólipos planctónicos

SISTEMÁTICA (Pugh, 1999)

Phylum CNIDARIA

Subphylum Medusozoa

Clase Hydrozoa

Subclase Hidroidolina

Orden Siphonophorae

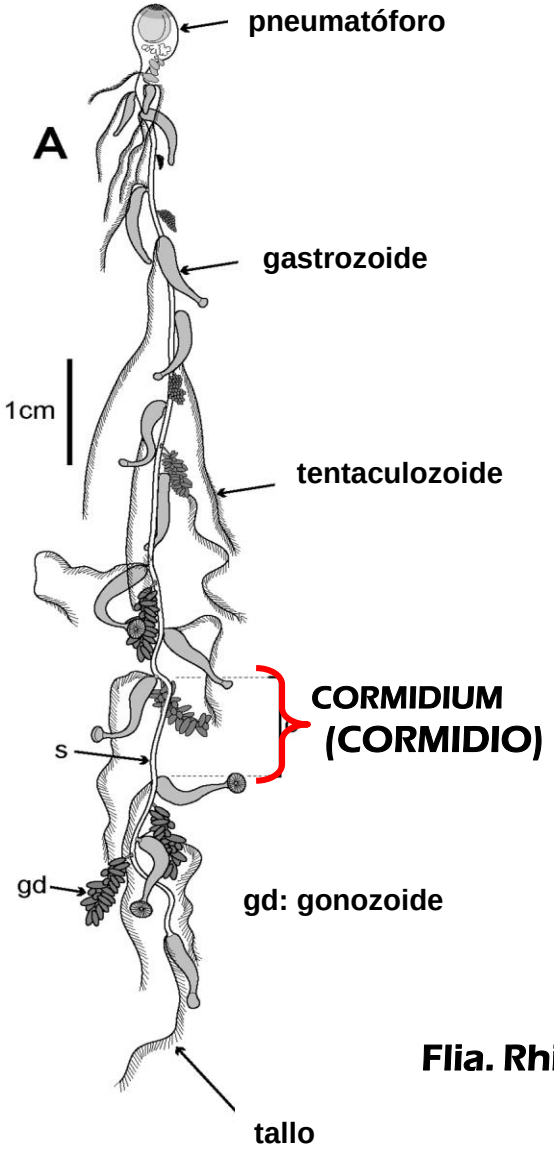
SubOrden Cystonectae (5 spp)

Flia Physaliidae: *Physalia physalis.*

Flia. Rhizophysidae: *Rhizophysa* sp.
Nanomia sp.



CYSTONECTAE

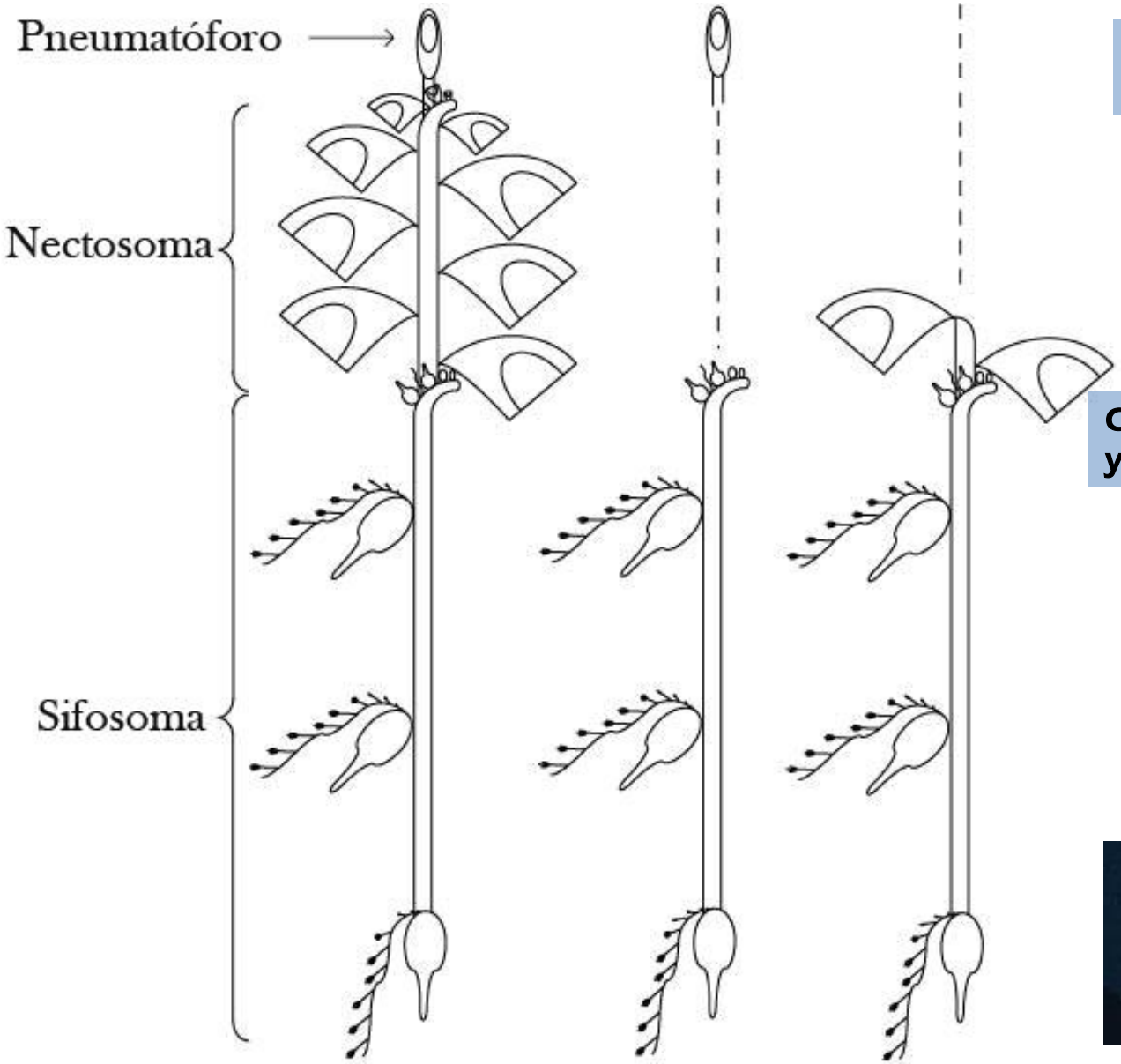






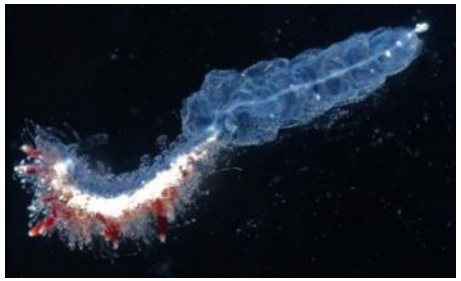
Flia. Rhizophysidae: *Rhizophysa* sp.
Nanomia bijuga

Flia Physaliidae: *Physalia physalis.*

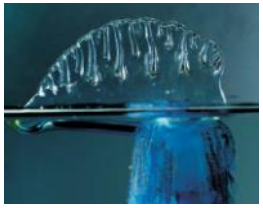


Suborden: Fisonéctidos Cystonéctidos Calycofóros

Fisonéctidos- pneumatóforo, nectosoma y sifosoma.



Cystonéctidos- pneumatóforo y sifosoma.



Calycóforos- nectosoma y sifosoma.

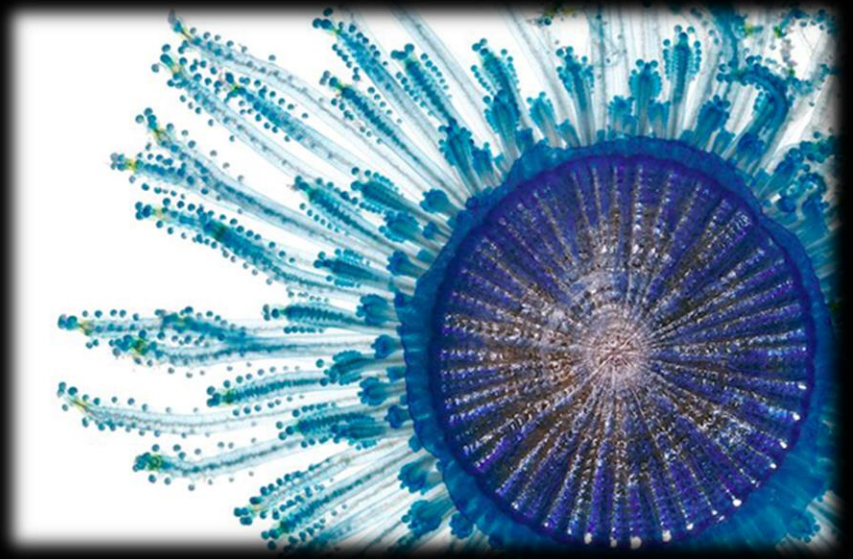


PATRÓN BÁSICO

Hydrozoa

Porpita porpita







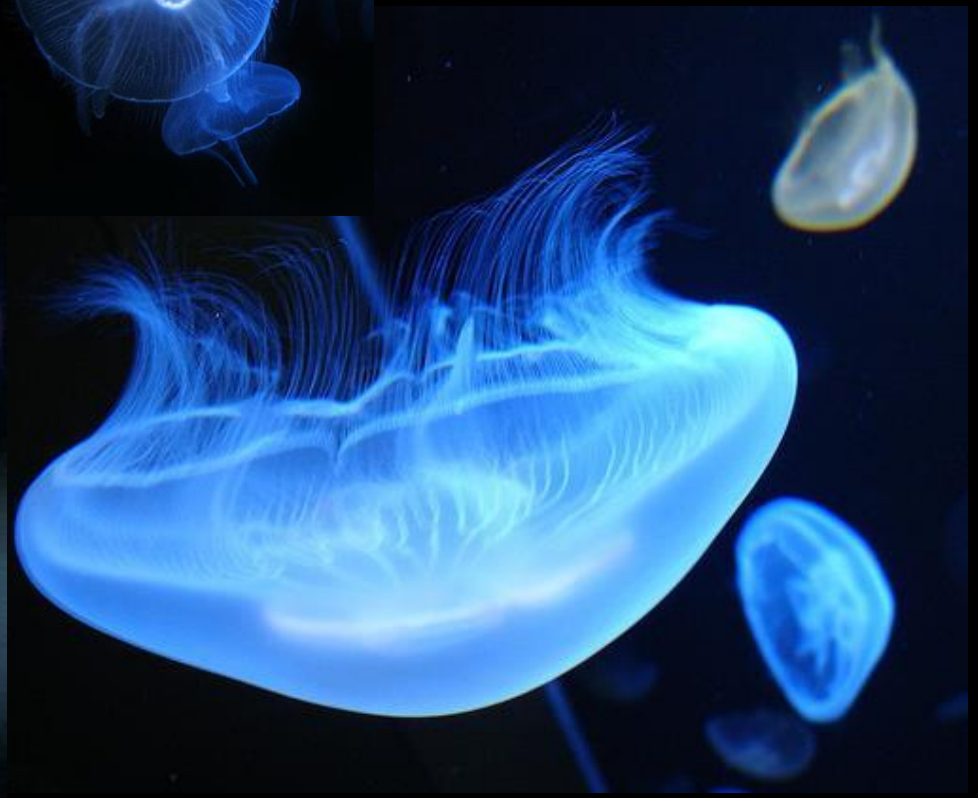


Velella velella

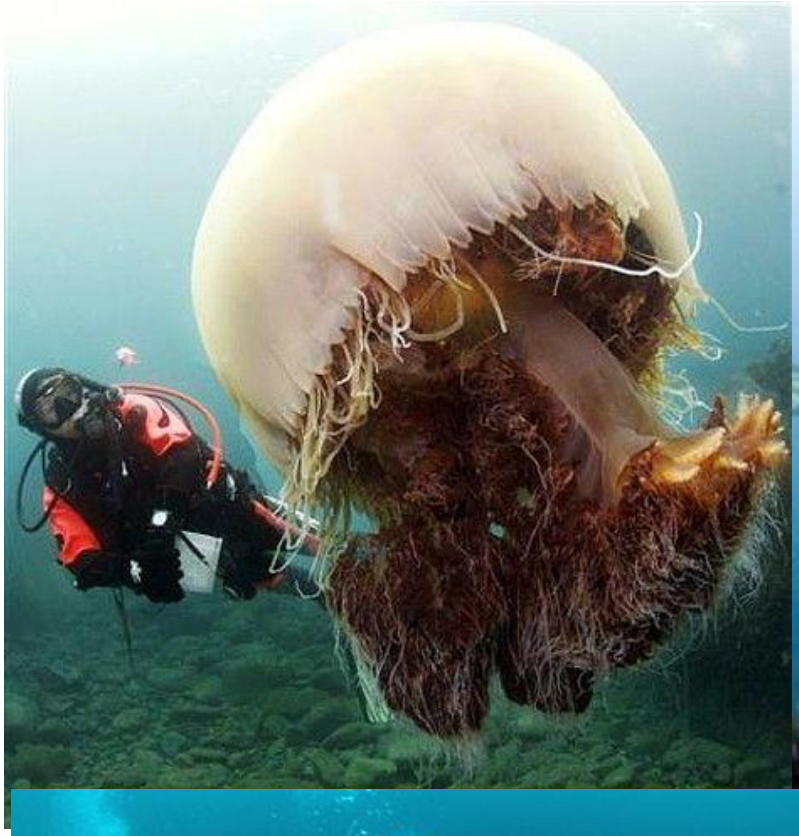
CLASE SCHYPHOZOA







Orden: Rhizostomeae. ***Nemopilema nomurai*** (medusa nomurai)







Cyanea capillata (**Medusa melena de león**) - Oc. Atlántico N.

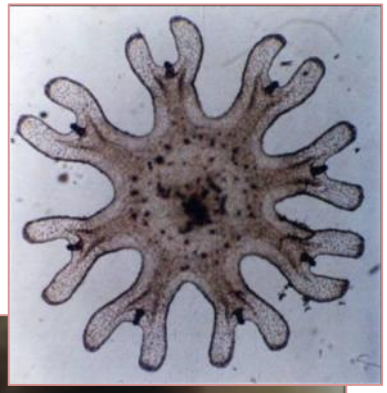


•Order Semaeostomeae
•Family Cyaneidae

- tentáculos en 8 grupos de 150 c/u

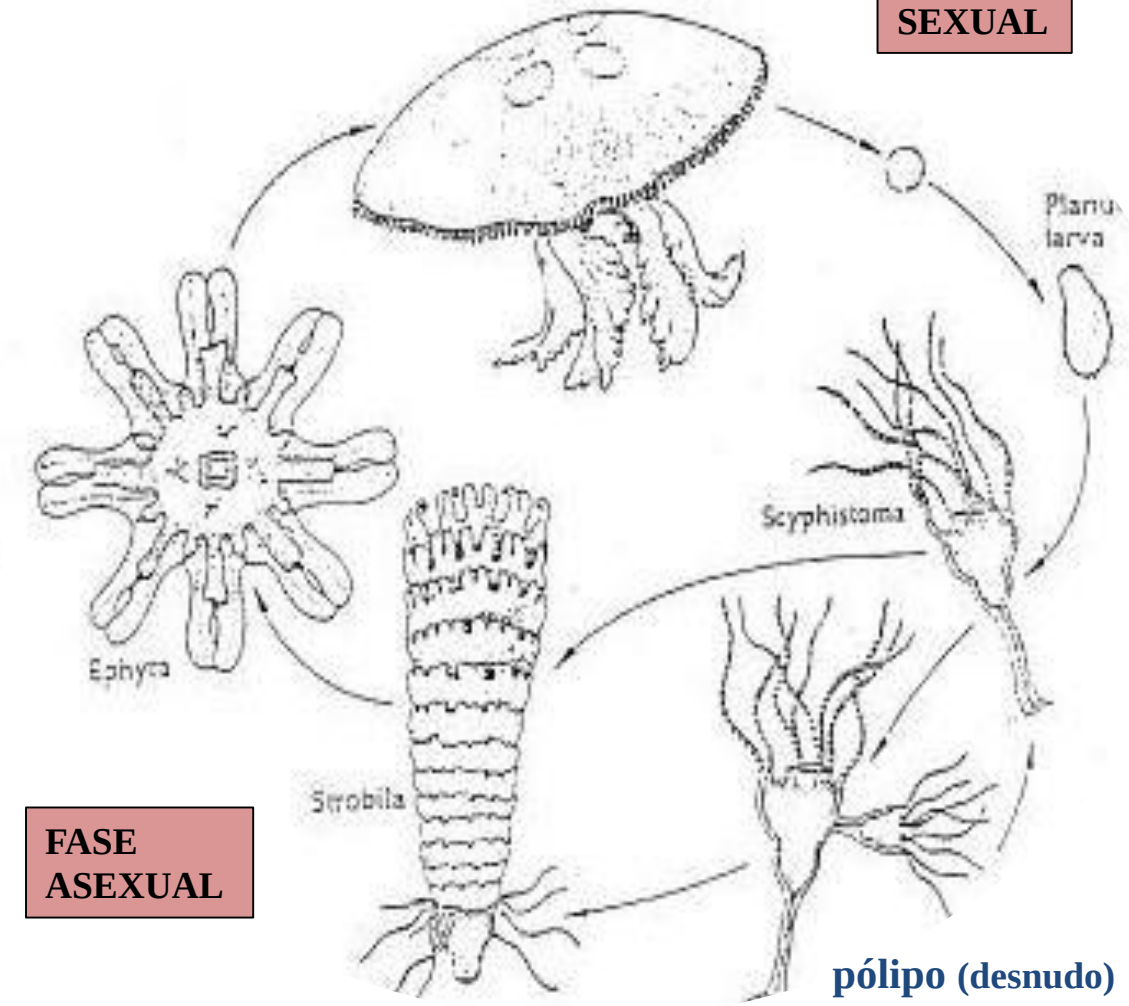
- elevada toxicidad

- más de 2 m de diámetro de campana





FASE SEXUAL



FASE ASEXUAL

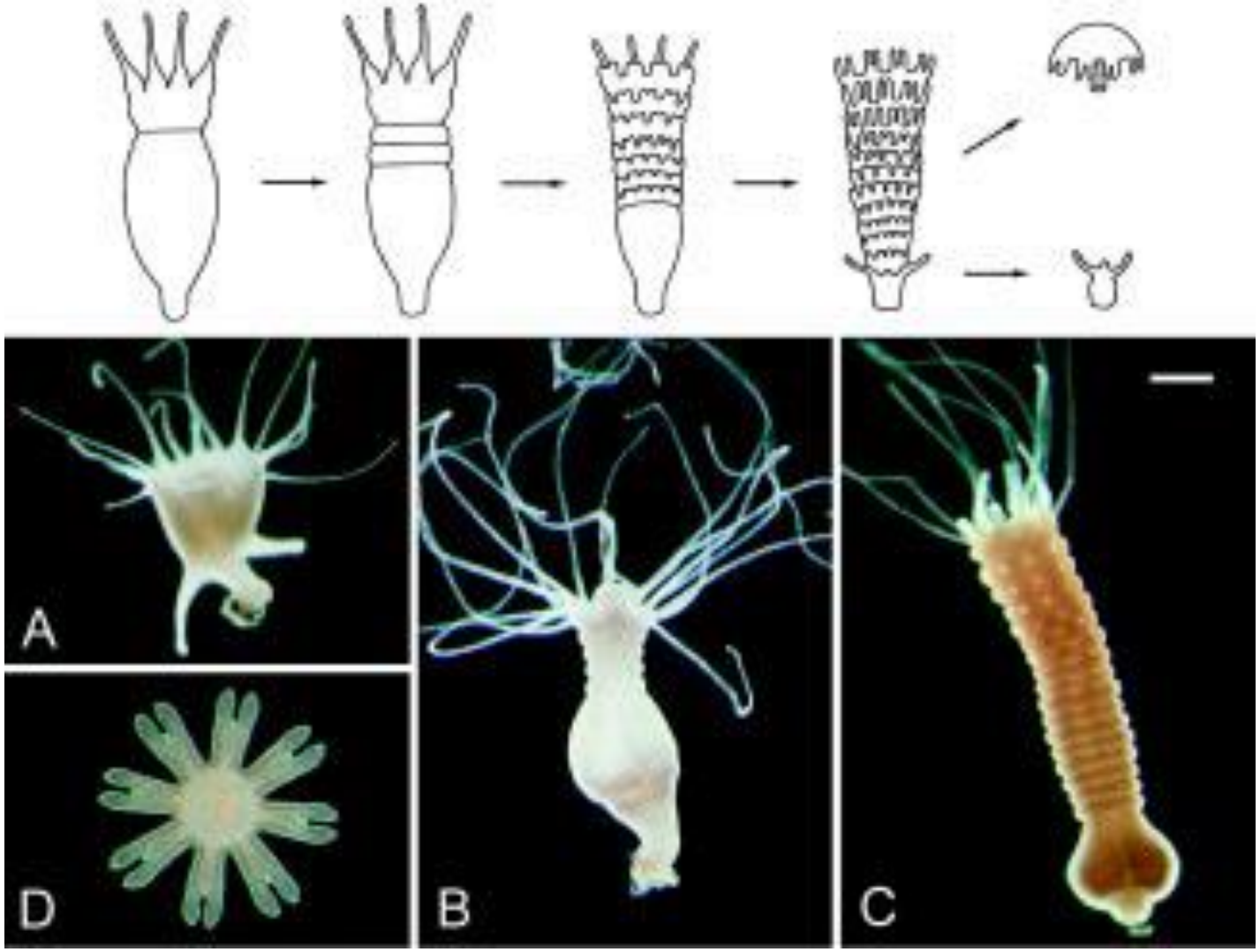
pólipo (desnudo)

Estrobilación: 1) Monodisco
2) Polidisco

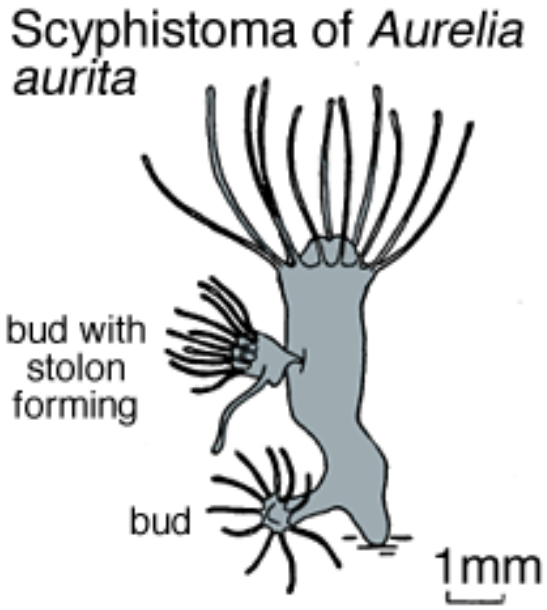
- **brotación**
- **fisión transversal**



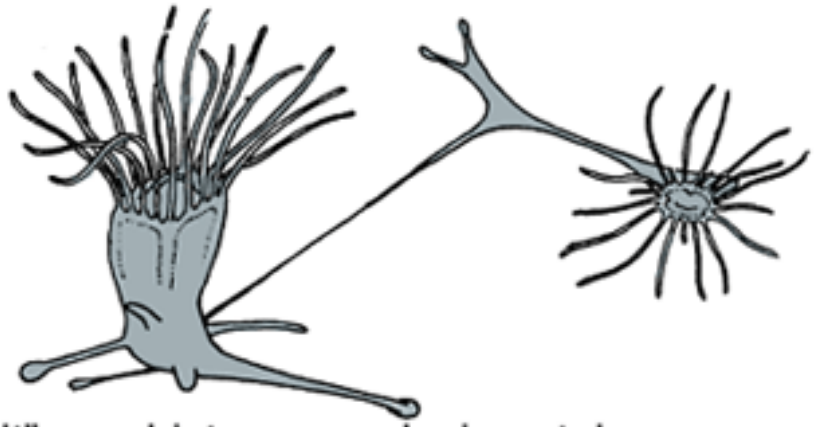



A
D
B
C

1

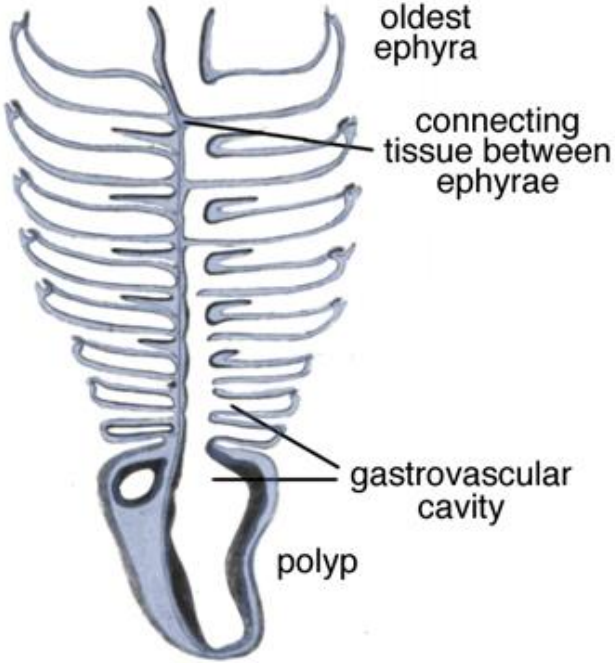


2



"Adult" scyphistoma producing stolons and new polyp buds

3

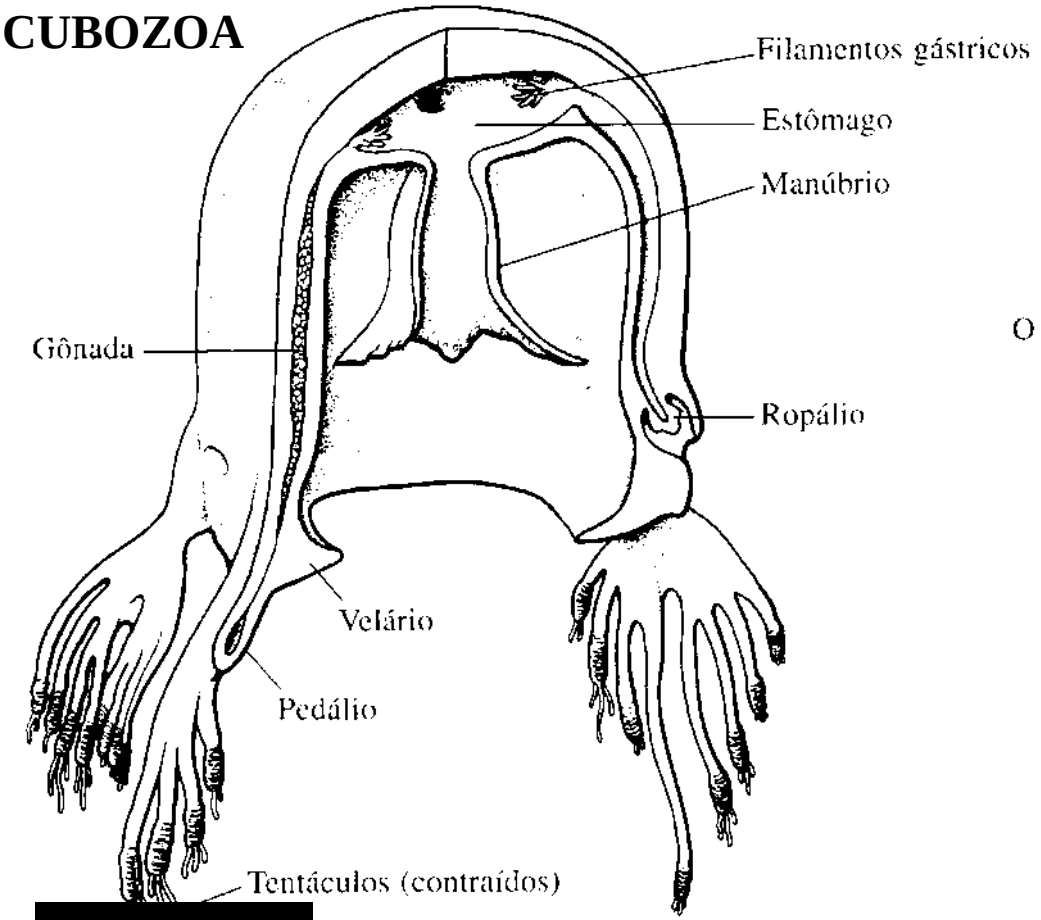


Scyphistoma estrobilando
(corte longt.)

CUBOZOA



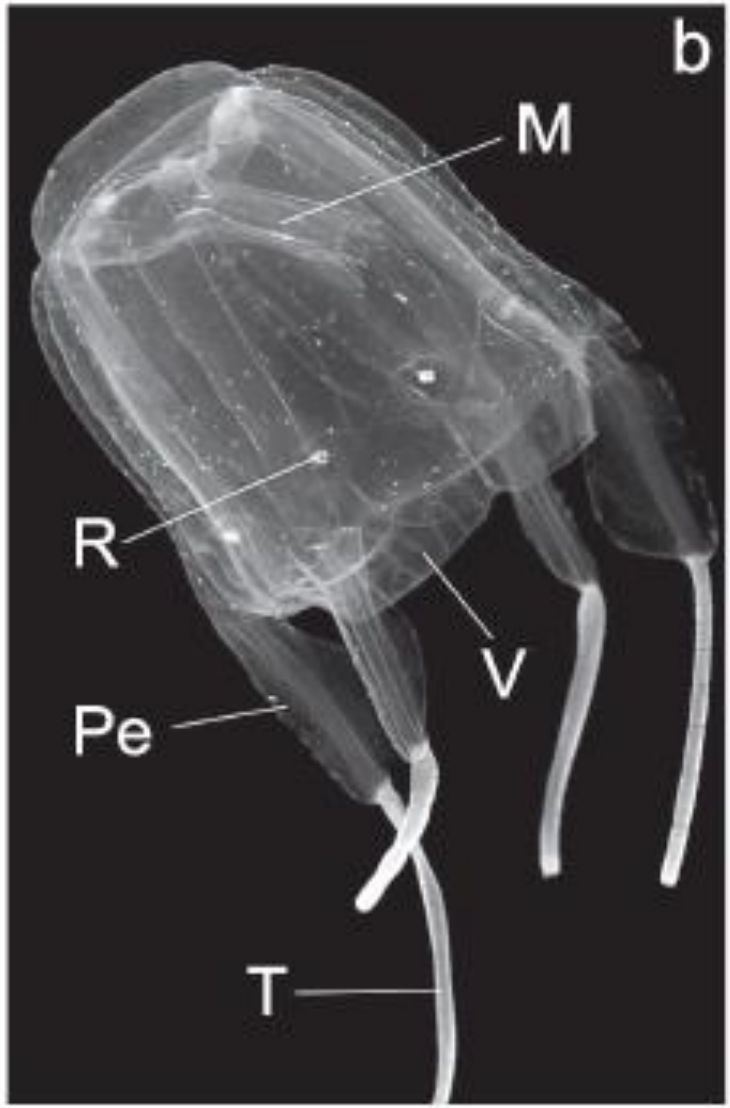
CUBOZOA



O



«Avispa de mar»: potente veneno neurotóxico



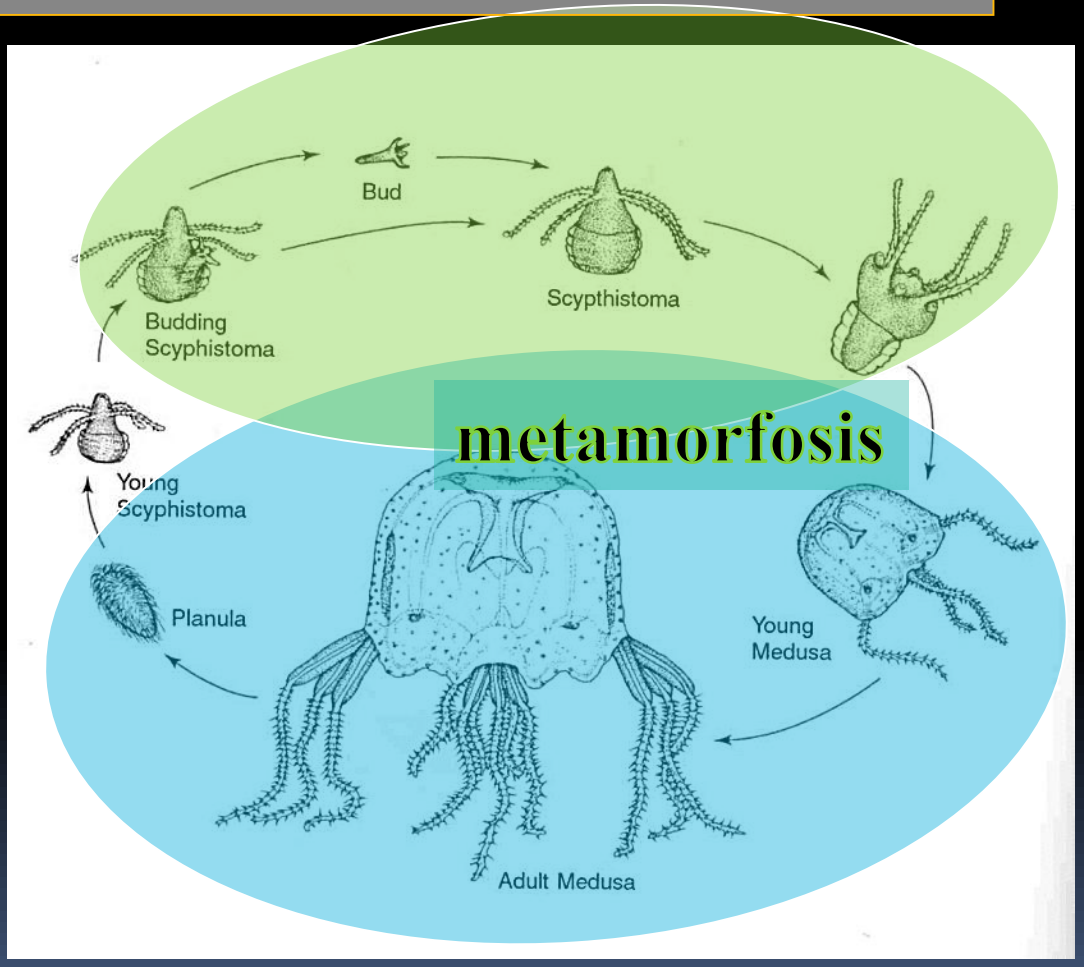
'a. T: tentáculos; R: ropalios; Pe: pedalios; M: manubrio; V: velarium.

Tamoya sp

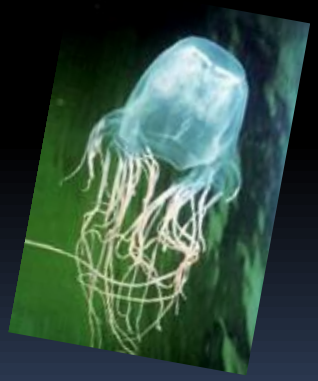


Cubozoa

ASEXUAL = POLIPO = BENTOS



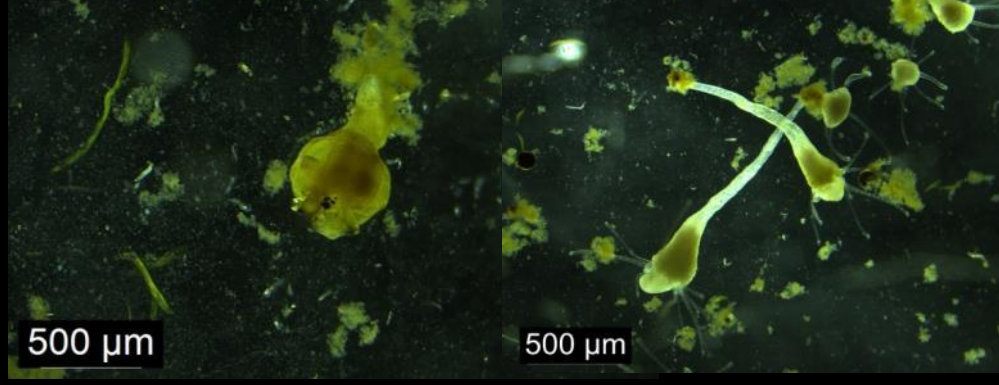


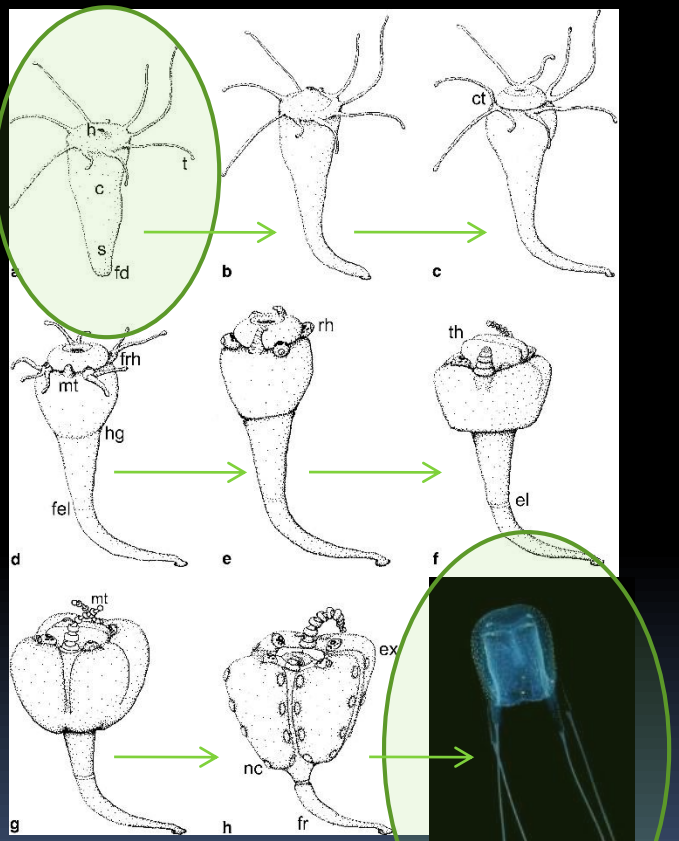


SEXUAL = MEDUSA = PLANCTON

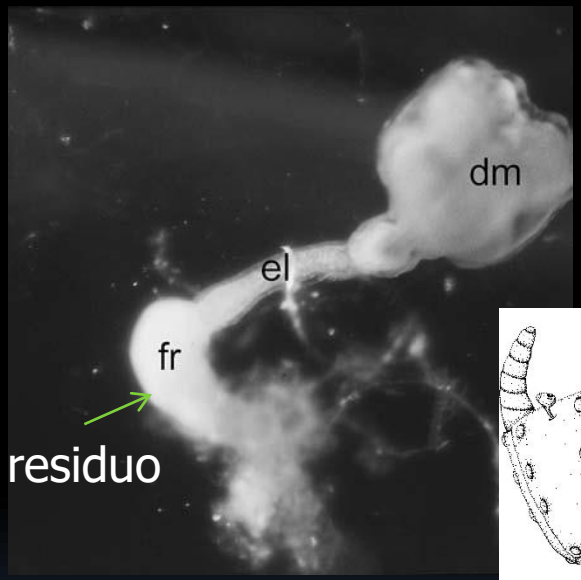
Carybdea marsupialis

2 tipos de metamorfosis:



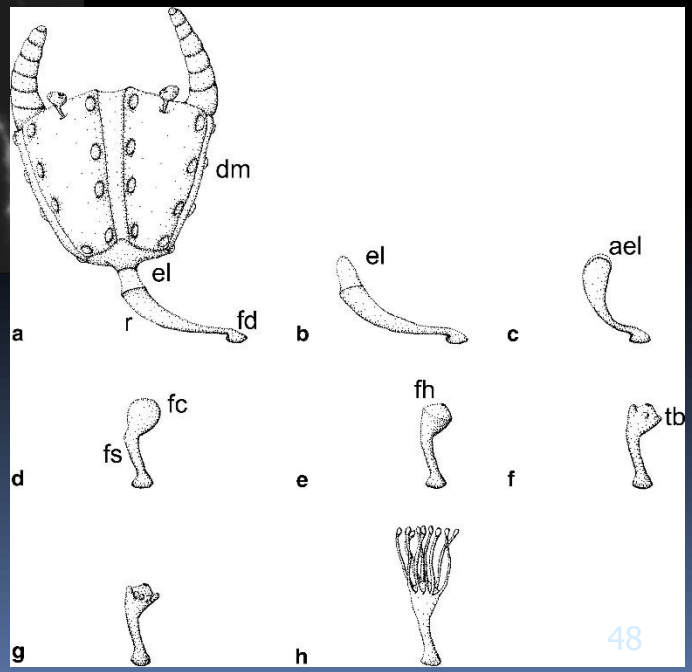


normal



Straehler-Pohl & Jarms (2005)

excepcional



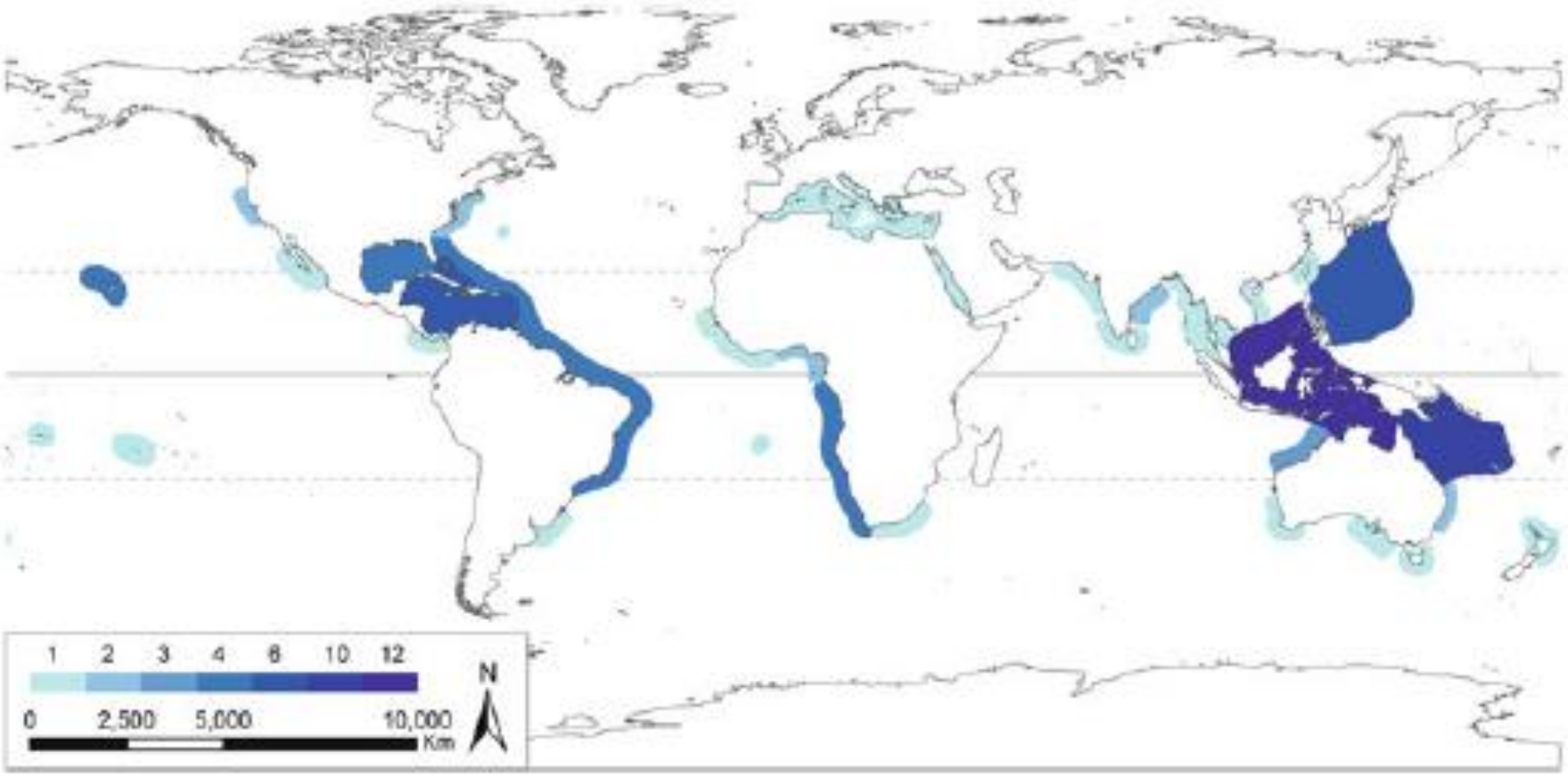
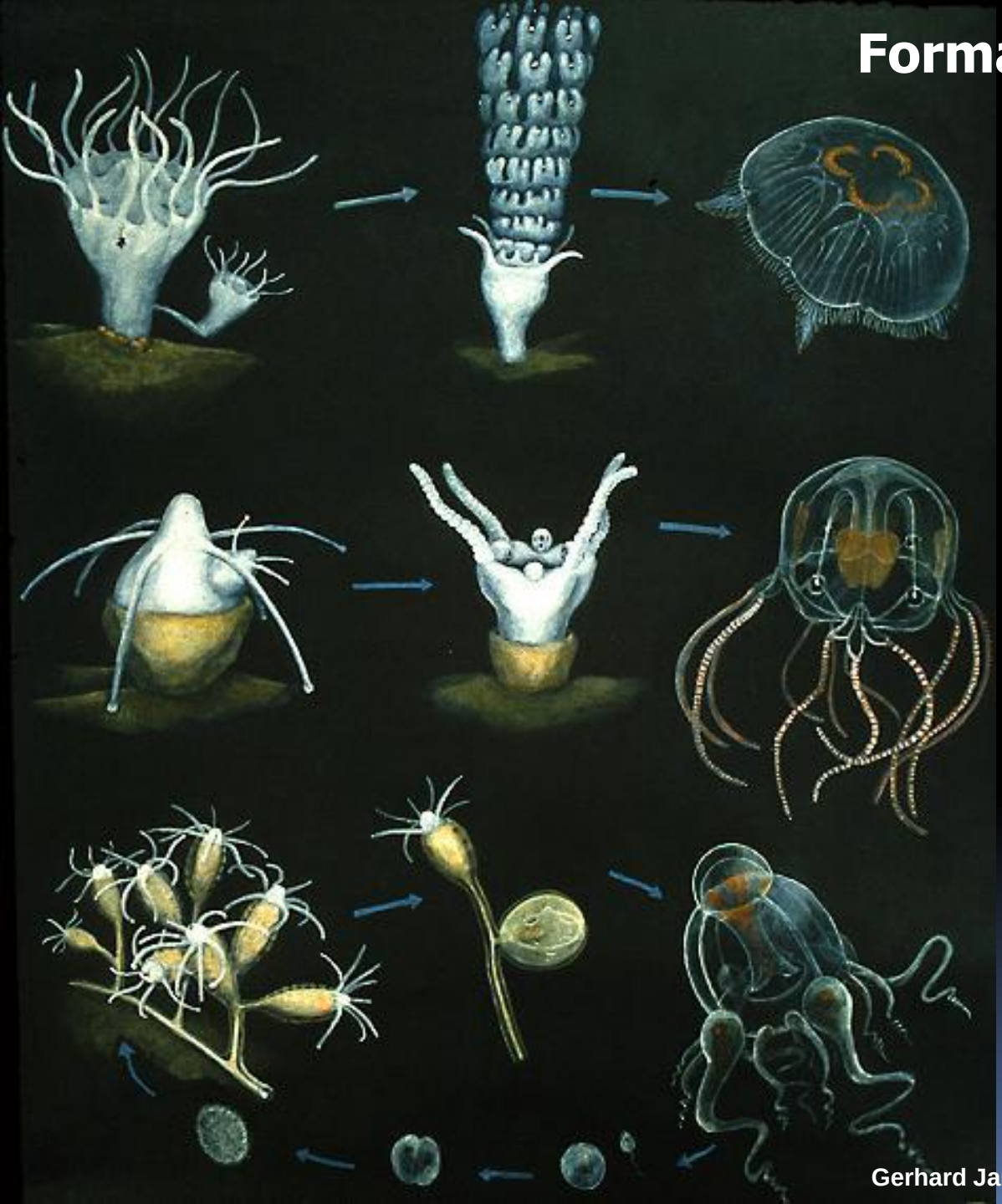


Fig. 12.3 Published occurrences of accepted cubozoan species around the world. *Colour* indicates number of accepted species for region. The width of regional patches is for presentation only; the majority of taxa were caught close to shore. For detail of species occurrence see Table 12.1

Formación de las medusas



Scyphozoa

Cubozoa

Hydrozoa

Gerhard Jarms (~2000)

Cubozoa



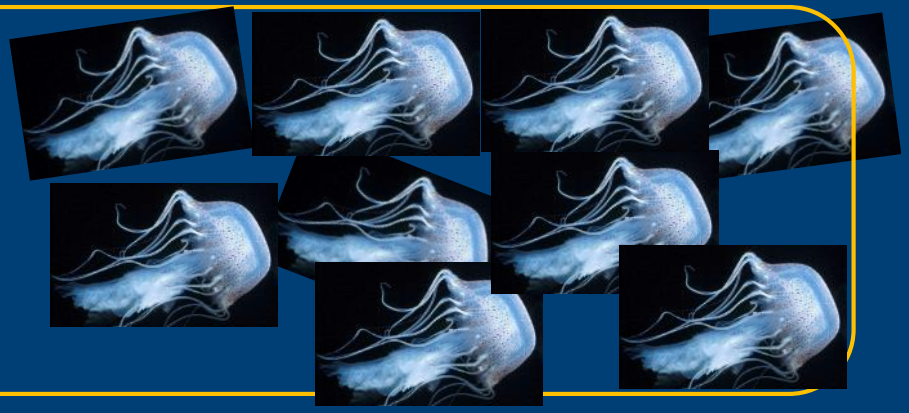
metamorfosis



Scyphozoa



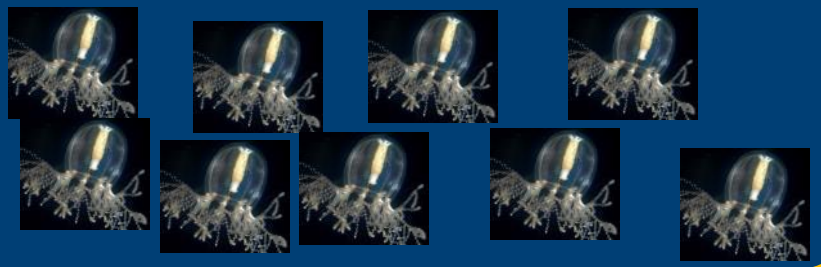
estrobilación



Hydrozoa



brotación



CLASE: STAUROZOA (Marques y Collins, 2004)

- 50 spp descriptas

(Miranda et al.,2017)

Clase: Staurozoa

Ciclo holobentónico

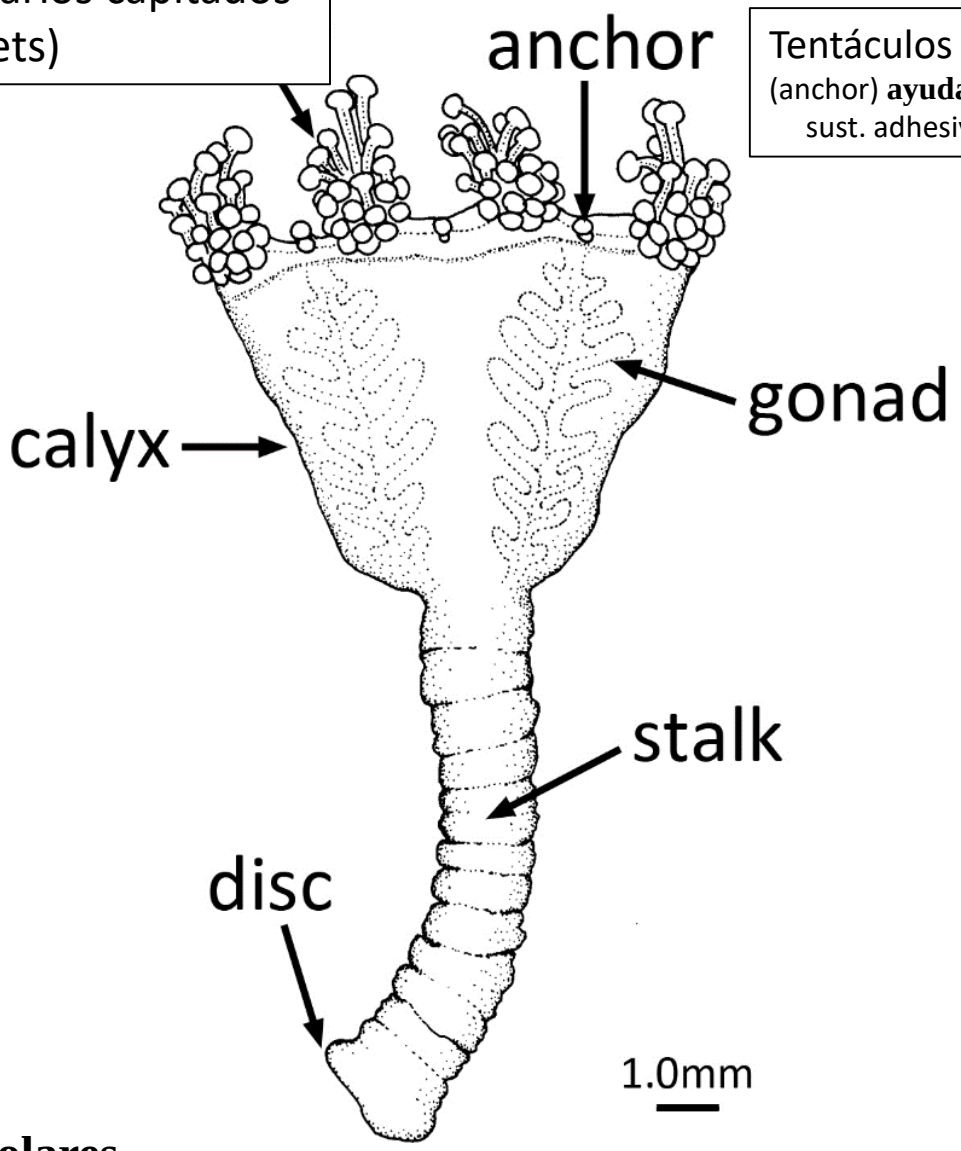


- aunque mantienen la fase medusoide, han reducido al máximo su vida pelágica

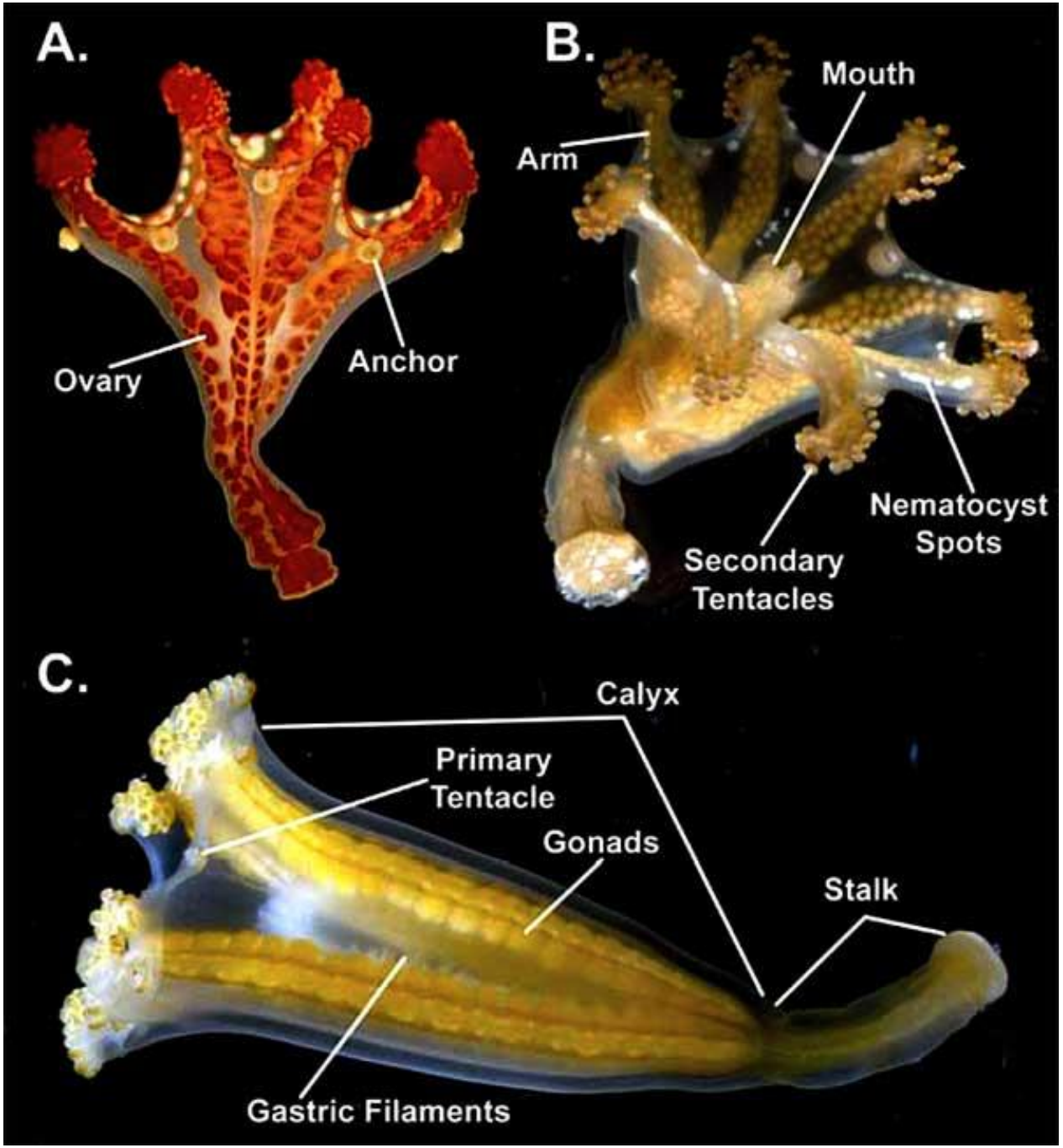
CLASE: STAUROZOA (Marques y Collins, 2004)

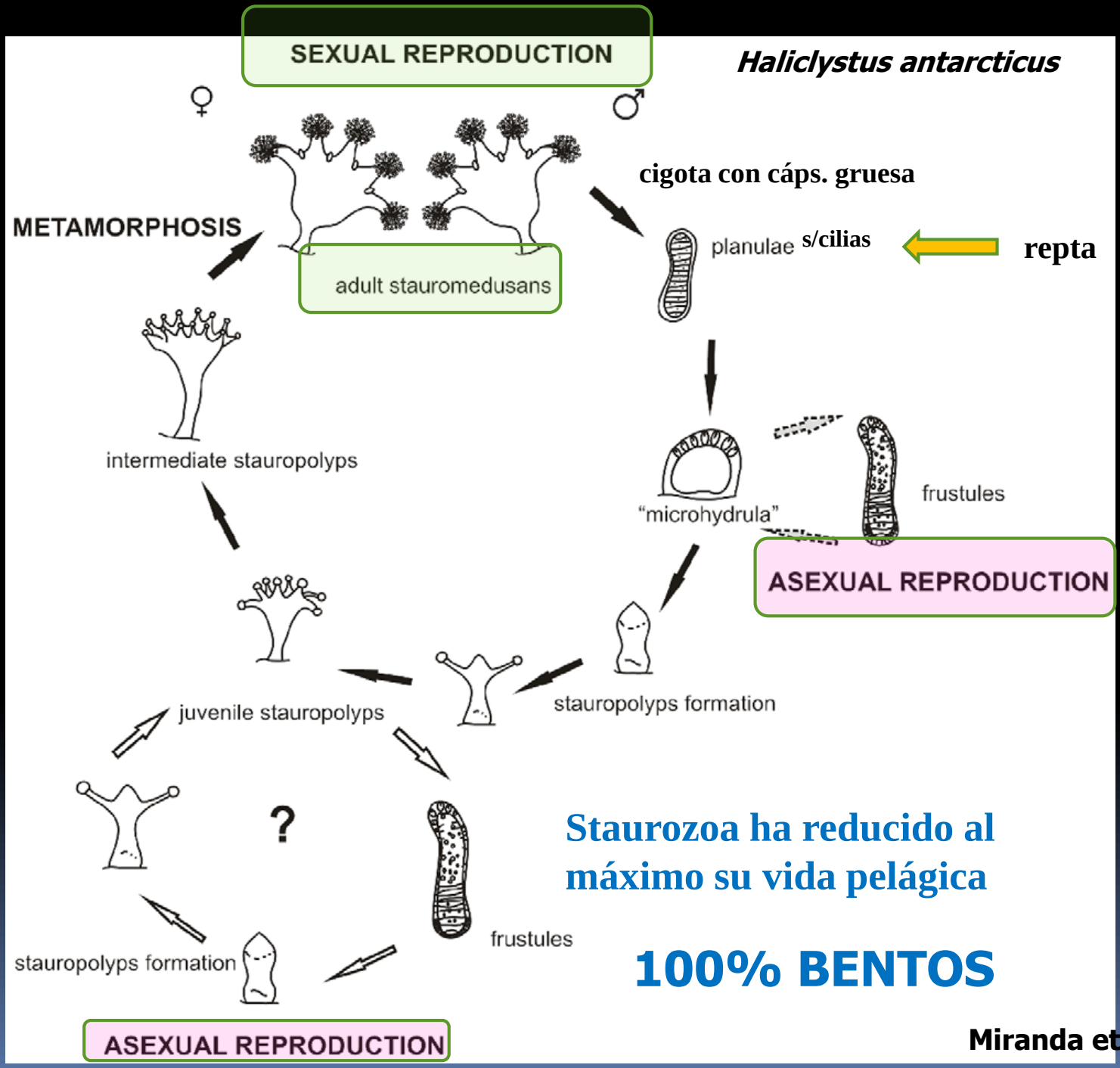
- Simetria tetrámera
  (= Cubozoa)

-



- Aguas tropicales, templadas y polares


A.
Ovary
Anchor
B.
Mouth
Arm
Nematocyst
Spots
Secondary
Tentacles
C.
Calyx
Primary
Tentacle
Gonads
Stalk
Gastric Filaments


SEXUAL REPRODUCTION
Haliclystus antarcticus
♀
♂
cigota con cáps. gruesa
METAMORPHOSIS
adult stauromedusans
planulae s/cilias
repta
intermediate stauropolyps
"microhydrula"
frustules
ASEXUAL REPRODUCTION
juvenile stauropolyps
stauropolyps formation
?
frustules
stauropolyps formation
ASEXUAL REPRODUCTION
Staurozoa ha reducido al máximo su vida pelágica
100% BENTOS
Miranda et al 2010

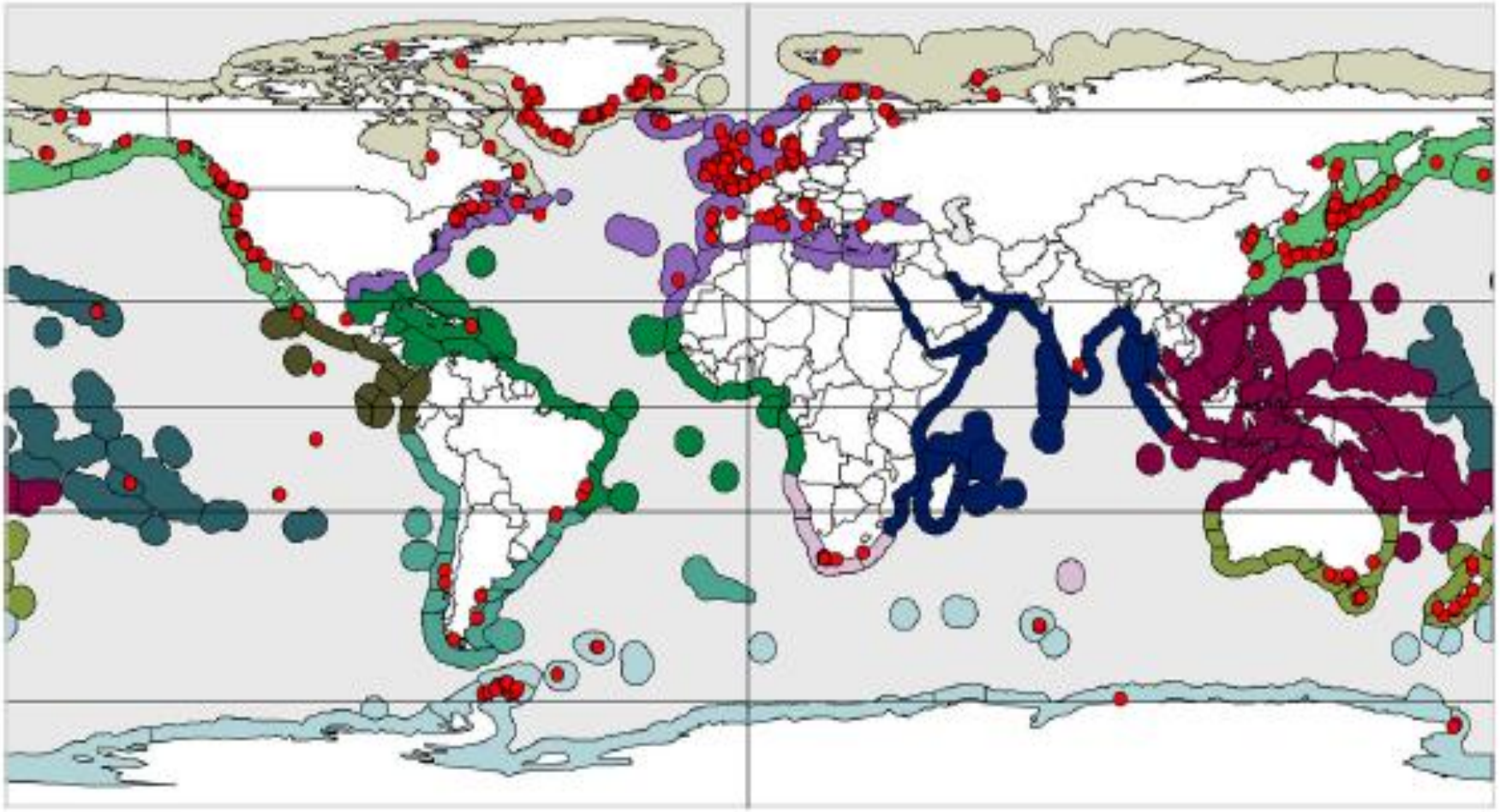


Fig. 4 Geographic distribution of Staurozoa. Distribution (*filled red circles*) plotted over the marine realms proposed by Spalding et al. (2007)

Disribución geográfica de Staurozoa (puntos rojos)